Can Time Series Foundation Models Forecast Realized Volatility?

Alessio Brini*

March 13, 2026

Abstract

Realized volatility forecasting is central to risk management and derivative pricing, yet no study has systematically assessed whether pretrained time series foundation models (TSFMs) can match established econometric benchmarks for this task. We evaluate eight zero-shot TSFMs against six econometric specifications on 50 assets from the VOLARE dataset (equities, FX, futures; 2015–2026) at horizons of 1, 5, and 22 days, using Diebold–Mariano tests and Model Confidence Sets. The best TSFMs outperform the best econometric benchmarks across all horizons and asset classes. Sundial, a flow-matching generative model, achieves the lowest quasi-likelihood (QLIKE) loss at every horizon and enters the Model Confidence Set for 97.5–100% of equities; its advantage over the best econometric specification (Log-HAR) grows from 11% at $h = 1$ to 37% at $h = 22$. However, performance is highly heterogeneous across architectures: some models produce severely miscalibrated forecasts, and model selection within the TSFM class matters as much as the choice between TSFMs and econometric benchmarks. These results suggest that practitioners can deploy zero-shot TSFMs as a drop-in replacement for HAR-family models, provided they evaluate multiple architectures rather than adopting a single TSFM by default.

*Duke University. Email: `alessio.brini@duke.edu`.

Keywords: Realized volatility, heterogeneous autoregressive model, time series foundation models, zero-shot forecasting

JEL Codes: C22, C53, C55, G17

1 Introduction

Forecasting realized volatility is central to risk management, derivative pricing, and portfolio allocation. Since the work of Andersen and Bollerslev (1998) and Barndorff-Nielsen and Shephard (2002), realized volatility constructed from high-frequency returns has become the standard model-free measure of ex-post price variation. The Heterogeneous Autoregressive (HAR) model of Corsi (2009) is the dominant forecasting benchmark: its three-component structure, aggregating past realized volatility at daily, weekly, and monthly frequencies, provides a parsimonious approximation to the long-memory dynamics that characterize volatility. Extensions incorporating jump variation (Andersen et al., 2007), semivariance asymmetries (Patton and Sheppard, 2015), and measurement error corrections (Bollerslev et al., 2016) have yielded mixed practical gains (Clements and Preve, 2021; Branco et al., 2024). Machine learning methods, including neural networks (Bucci, 2020; Brini and Toscano, 2025), random forests (Luong and Dokuchaev, 2018), graph-based approaches (Zhang et al., 2024, 2025), and convolutional architectures (Moreno-Pino and Zohren, 2024), have also been applied, but a broader assessment by Christensen et al. (2023) concludes that machine learning offers incremental improvements rather than a paradigm shift.

A different class of models has emerged over the past two years: time series foundation models (TSFMs). These are large pretrained transformers (Vaswani et al., 2017) trained on billions of time series observations across domains, capable of producing forecasts for previously unseen series without any retraining, a capability known as zero-shot forecasting. Leading examples include Chronos (Ansari et al., 2024), Moirai (Woo et al., 2024), and Lag-Llama (Rasul et al., 2024), several of which have released second-generation versions with improved architectures (Ansari et al., 2025; Liu et al., 2025a). Surveys document the rapid

pace of development in this area (Liang et al., 2024; Ye et al., 2024; Miller et al., 2024). Outside finance, Carriero et al. (2024) applied zero-shot language models to macroeconomic forecasting and found that they do not outperform traditional econometric baselines, raising the question of whether similar conclusions hold for financial time series.

Applications of TSFMs to financial time series remain limited. Goel et al. (2025) tested TimesFM 1.0 on realized volatility for 20 stocks and found that fine-tuning was necessary for the model to compete with HAR; zero-shot performance was poor. However, that negative result may be model-vintage-specific rather than a fundamental limitation of zero-shot inference: TimesFM 2.5, released one year later, performs well zero-shot in our evaluation. Rahimikia et al. (2025) evaluated several TSFMs on daily stock returns and reported uniformly negative zero-shot results. Realized volatility differs from returns in ways that may favor foundation models: it is strictly positive, exhibits well-documented mean reversion and long-memory dependence (Andersen et al., 2003), and follows a distribution closer to the macroeconomic and physical series on which TSFMs were pretrained.

To our knowledge, no study has yet evaluated multiple foundation models on realized volatility with formal statistical testing. This paper conducts the first systematic comparison of zero-shot TSFMs against established econometric benchmarks for realized volatility forecasting. We evaluate eight TSFMs—spanning six distinct architectures—against six econometric specifications across 50 assets spanning three asset classes (equities, FX, futures) and three forecast horizons ($h = 1, 5, 22$ days), using the recently released VOLARE dataset (Cipollini et al., 2026). The zero-shot setting, with no domain-specific training, isolates the value of general time series pretraining. We apply Diebold–Mariano (DM) tests (Diebold and Mariano, 1995) and Model Confidence Set (MCS) analysis (Hansen et al., 2011) to control for multiple comparisons.

The best zero-shot TSFMs outperform the best econometric benchmarks across all horizons and asset classes, contrasting with prior negative findings on daily returns and on TimesFM 1.0. Sundial, a flow-matching generative model, achieves the lowest QLIKE at

every horizon and enters the MCS for 97.5–100% of equities. The TSFM advantage grows with forecast horizon, from 11% at $h = 1$ to 37% at $h = 22$. However, performance is highly heterogeneous across TSFM architectures: some models produce severely miscalibrated forecasts while others outperform all benchmarks, indicating that model selection within the TSFM class is as important as the choice between TSFMs and econometric models.

The remainder of this paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the VOLARE dataset. Section 4 presents the econometric and foundation model specifications, along with the forecast evaluation framework. Section 5 reports the empirical results on univariate realized variance. Section 6 assesses statistical significance through DM tests, MCS, Mincer–Zarnowitz regressions, and Giacomini–Rossi fluctuation tests. Section 7 contains robustness checks, and Section 8 concludes.

2 Literature Review

2.1 Realized Volatility Modeling

The theory of realized volatility originates with Andersen and Bollerslev (1998) and Andersen et al. (2001), who showed that the sum of squared intraday returns provides a consistent, nonparametric estimator of the integrated variance of asset prices. Barndorff-Nielsen and Shephard (2002) formalized this result in a semimartingale framework, establishing that realized variance converges to integrated variance as the sampling frequency increases. Subsequent work identified the key stylized facts of realized volatility: approximate log-normality, long-memory dependence with a fractional integration parameter $d \approx 0.4$, and slow mean reversion (Andersen et al., 2003).

At ultra-high sampling frequencies, microstructure noise from bid-ask bounce and discrete price changes biases the realized variance estimator upward. Barndorff-Nielsen et al. (2008) introduced the realized kernel as a noise-consistent alternative that remains valid in the presence of market microstructure effects.

The HAR model of Corsi (2009) became the standard forecasting benchmark for realized volatility. By including daily, weekly, and monthly RV components as regressors, the model approximates long-memory behavior through a cascade of time scales corresponding to heterogeneous trading horizons. The specification is parsimonious (three regressors plus a constant) yet achieves accuracy comparable to far more complex alternatives.

A large body of work has extended the HAR framework. The HAR-J model (Andersen et al., 2007) separates realized variance into continuous and jump components using bipower variation (Barndorff-Nielsen and Shephard, 2004). The HAR-RS model (Patton and Sheppard, 2015; Barndorff-Nielsen et al., 2010) decomposes realized variance into positive and negative semivariance to capture asymmetric responses to upside and downside moves. The HARQ model (Bollerslev et al., 2016) interacts the daily RV regressor with realized quarticity to account for time-varying measurement error. Clements and Preve (2021) provides a practical guide to implementing and comparing these extensions.

Long-memory models offer a complementary approach. Autoregressive Fractionally Integrated Moving Average (ARFIMA) models (Baillie et al., 1996; Izzeldin et al., 2019) explicitly parameterize the fractional integration order and remain competitive at longer forecast horizons where the slow mean reversion of realized volatility becomes the dominant dynamic.

Machine learning methods have been applied to realized volatility forecasting with mixed results. Bucci (2020) reports that feedforward and recurrent neural networks outperform HAR on a sample of U.S. stocks. Luong and Dokuchaev (2018) finds gains from random forests. Moreno-Pino and Zohren (2024) introduces DeepVol, a convolutional neural network that processes raw order book data. In contrast, Branco et al. (2024) concludes that simple linear models are difficult to beat after correcting for multiple testing, and Christensen et al. (2023) finds that machine learning offers only incremental improvements over standard econometric methods. Recent surveys by Gunnarsson et al. (2024) and Leushuis and Petkov (2026) cover this literature.

2.2 Time Series Foundation Models

Time series foundation models are large pretrained models that produce forecasts for arbitrary time series without task-specific training, analogous to large language models for text (Liang et al., 2024). TSFMs differ from task-specific deep learning models (neural networks, recurrent networks, convolutional architectures) in a fundamental way: deep learning models must be trained from scratch on the target series before they can produce any forecast, while TSFMs are pretrained on large external corpora and can forecast immediately via zero-shot inference. This eliminates the need for domain-specific training data, hyperparameter tuning, and GPU-intensive estimation—a practical advantage when deploying to new assets or markets where historical data may be limited. TSFMs are trained on vast corpora of time series drawn from diverse domains (weather, energy, retail, transport, macroeconomics) and learn general temporal patterns that transfer to unseen series at inference time.

Three architectural families dominate the first generation of TSFMs. Chronos (Ansari et al., 2024) converts continuous values into discrete tokens via quantile binning and uses a T5 encoder-decoder architecture for autoregressive forecasting. T5, the Text-to-Text Transfer Transformer (Raffel et al., 2020), is a sequence-to-sequence model that reads the full input through an encoder and generates output tokens autoregressively through a decoder. Moirai (Woo et al., 2024) handles arbitrary numbers of variates through an "any-variate" attention mechanism and uses mixture distribution outputs for uncertainty quantification. Lag-Llama (Rasul et al., 2024) adapts the LLaMA decoder-only architecture for probabilistic time series forecasting using lag-based tokenization. Finance-specific foundation models have also been proposed, including Kronos (Shi et al., 2025), which is pretrained on candlestick (open-high-low-close-volume, OHLCV) data from over 45 global exchanges using a learned tokenizer that maps price patterns into discrete tokens.[1]

Second-generation models appeared in 2025. Ansari et al. (2025) introduced Chronos-

[1]We do not evaluate Kronos because it expects OHLCV price data as input and cannot be meaningfully applied to univariate realized variance without constructing synthetic candlestick data, which introduces an arbitrary input transformation unrelated to the model's intended use case.

2 and Chronos-Bolt: the former adds multivariate support and covariate handling, while Chronos-Bolt is a distilled variant, meaning its parameters were trained to approximate the full Chronos-2 model's behavior at a fraction of the computational cost, producing direct quantile forecasts in a single forward pass and running up to 250 times faster than the original. Moirai 2.0 (Liu et al., 2025a) demonstrated that smaller, better-trained models can match or exceed their larger predecessors using multi-token prediction and improved tokenization. Other models include TimesFM (Das et al., 2024), a patch-based decoder model from Google; Toto (Cohen et al., 2024), an observability-focused model from Datadog; Moirai-MoE (Liu et al., 2024a), a sparse Mixture of Experts extension of Moirai; Sundial (Liu et al., 2025b), which uses flow matching for generative forecasting; MOMENT (Goswami et al., 2024), a masked-encoder architecture designed for multiple time series tasks (forecasting, classification, anomaly detection); and Timer (Liu et al., 2024b).

Standardized benchmarks have been developed to evaluate TSFMs across domains. GIFT-Eval (Aksu et al., 2024) provides a unified evaluation protocol across multiple datasets and forecast horizons. FEV-Bench (Shchur et al., 2025) focuses on forecasting evaluation with emphasis on calibration. TSFM-Bench (Li et al., 2024) compares models across zero-shot and fine-tuned settings. A consistent finding is that no single model dominates across all domains and horizons, which motivates domain-specific evaluations such as the one we undertake here. Adler et al. (2025) raises additional concerns about the calibration of TSFM prediction intervals, finding systematic miscoverage across several models.

2.3 Foundation Models for Volatility Forecasting

The application of TSFMs to volatility forecasting is recent and limited. The closest prior work is Goel et al. (2025), who tested TimesFM 1.0 on realized volatility for 20 stocks. They found that zero-shot TimesFM underperformed HAR and that fine-tuning was necessary to achieve competitive accuracy. Our paper differs in several respects: we evaluate eight foundation models spanning six distinct architectures (Chronos-Bolt, Moirai 2.0 and Moirai-

MoE, Lag-Llama, TimesFM 2.5, Toto, and Sundial), use newer model generations, and apply MCS analysis to control for multiple comparisons. The same research group also applied TimesFM to Value-at-Risk forecasting (Goel et al., 2024).

Rahimikia et al. (2025) evaluated Chronos, TimesFM, Moirai, and MOMENT on daily stock returns and found that zero-shot TSFMs consistently underperformed ARIMA. However, daily returns differ from realized volatility in fundamental ways: returns are approximately symmetric, near-white-noise, and lack the strong serial dependence that characterizes realized volatility. Realized volatility is strictly positive, mean-reverting, and displays long-memory dynamics, properties that bring it closer to the macroeconomic and physical time series on which TSFMs were pretrained. This distinction suggests that negative results on returns need not carry over to realized measures.

Other financial applications include stock price forecasting (Łaniewski and Ślepaczuk, 2025; Valeyre and Aboura, 2024), foreign exchange volatility modeling (Nguyen et al., 2025), and multivariate financial time series (Marconi, 2025). Finance-specific foundation models have also been proposed, including FinCast (Zhu et al., 2025) and Kronos (Shi et al., 2025).

The question of whether to fine-tune or apply TSFMs zero-shot is itself an active area of research. Das et al. (2025) explored in-context fine-tuning for TimesFM, while Germán-Morales et al. (2024) and Gupta et al. (2024) investigated parameter-efficient adaptation via Low-Rank Adaptation (LoRA) (Hu et al., 2022). Testing zero-shot performance first establishes an important baseline: if a TSFM already performs well without adaptation, the marginal value of fine-tuning may be limited, and the zero-shot workflow offers practical advantages in simplicity and computational cost.

The gap in this literature—the absence of a multi-model, multi-asset evaluation with formal statistical testing for realized volatility—motivates the empirical design we describe next.

3 The VOLARE Dataset

We use the VOLARE (VOLatility Archive for Realized Estimates) dataset of Cipollini et al. (2026), which provides daily realized volatility measures for a broad cross-section of financial assets. VOLARE is constructed from ultra-high-frequency tick data sourced from Kibot, covering 40 U.S. equities, 5 major currency pairs, and 5 commodity and index futures contracts. The equity sample begins on January 2, 2015 and runs through January 30, 2026, yielding 2,786 trading days per stock. The FX and futures samples begin on September 25, 2009 (4,242 and 4,228 trading days, respectively).

For each asset-day, VOLARE provides realized measures computed at multiple sampling frequencies. All measures used in this paper are computed at the 5-minute sampling frequency: realized variance (RV), bipower variation (BPV), positive and negative realized semivariance (RS^+, RS^-), and realized quarticity (RQ). All values are expressed in decimal squared returns; for instance, a typical daily RV for a U.S. equity is approximately 1.9×10^{-4}, corresponding to an annualized volatility of roughly 22%. VOLARE is well suited for our multi-asset comparison because it constructs all realized measures using a uniform methodology across assets, ensuring that cross-asset differences in model rankings reflect genuine forecasting performance rather than measurement inconsistencies. Our equity sample comprises the 40 stocks in VOLARE that span the 2015–2026 period (approximately 2,786 trading days): AAPL, ADBE, AMD, AMGN, AMZN, AXP, BA, CAT, CRM, CSCO, CVX, DIS, GE, GOOGL, GS, HD, HON, IBM, JNJ, JPM, KO, MCD, META, MMM, MRK, MSFT, NFLX, NKE, NVDA, ORCL, PG, PM, SHW, TRV, TSLA, UNH, V, VZ, WMT, and XOM. These cover all 11 Global Industry Classification Standard (GICS) sectors. The FX sample consists of five major currency pairs (AUDUSD, EURUSD, GBPUSD, USDCAD, USDJPY) and the futures sample covers five contracts (Corn, Crude Oil, E-mini S&P 500, Gold, Natural Gas). Both FX and futures samples span 2009–2026 (approximately 4,240 trading days), providing substantially longer histories than the equity sample because VOLARE's intraday coverage for these asset classes begins earlier.

Table 1 reports descriptive statistics for the 5-minute realized variance across the three asset classes. As expected for a strictly positive quantity, realized variance is right-skewed across all asset classes, with heavy tails (kurtosis ranging from 35 to over 3,000) confirming that extreme volatility episodes are a persistent feature of the data. The first-order autocorrelation of daily RV, denoted ρ_1, averages 0.606 for equities, consistent with the well-documented persistence of volatility; the 22-day autocorrelation (ρ_{22}, i.e., the sample autocorrelation at lag 22) averages 0.085, indicating slow decay. The FX pairs display RV approximately two orders of magnitude smaller than equities (cross-sectional average $\bar{RV} \approx 3 \times 10^{-5}$ vs. 2×10^{-4}), with comparable persistence (cross-sectional average $\bar{\rho}_1 = 0.52$, where the bar denotes averaging across assets). Futures exhibit the widest cross-asset heterogeneity: the E-mini S&P 500 (ES) has $\rho_1 = 0.779$, while Gold (GC) shows near-zero autocorrelation ($\rho_1 = -0.001$), presenting a natural stress test for forecasting models.

Table 1: Descriptive statistics for daily realized variance measured at the 5-minute frequency. The equity panel reports cross-sectional averages across 40 stocks; FX and futures report individual assets. ρ_k denotes the sample autocorrelation at lag k (e.g., ρ_{22} is the autocorrelation at 22 trading days). All RV values are in decimal squared returns ($\times 10^{-4}$).

	N	Mean	Median	Skew	Kurt	ρ_1	ρ_{22}
Panel A: Equities (cross-sectional average, 40 stocks, 2015–2026)							
Average	2,786	1.91	1.12	13.5	269	0.606	0.085
Panel B: Foreign Exchange (5 pairs, 2009–2026)							
AUDUSD	4,241	0.48	0.37	11.3	232	0.577	0.138
EURUSD	4,241	0.28	0.22	5.0	51	0.570	0.263
GBPUSD	4,242	0.32	0.24	27.5	1,117	0.484	0.061
USDCAD	4,240	0.25	0.19	4.3	35	0.654	0.368
USDJPY	4,239	0.33	0.22	14.4	349	0.325	0.063
Panel C: Futures (5 contracts, 2009–2026)							
Corn (C)	4,185	2.48	1.54	12.3	238	0.178	0.079
Crude Oil (CL)	4,223	6.87	3.87	57.6	3,546	0.195	0.047
E-mini S&P (ES)	4,224	1.11	0.53	12.9	212	0.779	0.110
Gold (GC)	4,206	3.63	2.32	62.3	3,963	−0.001	−0.000
Natural Gas (NG)	4,221	10.30	6.35	19.0	522	0.187	0.267

4 Methodology

4.1 Econometric Benchmarks

We consider six econometric specifications that span the main approaches to realized volatility forecasting: the HAR family and its extensions, which exploit multi-scale aggregation, and ARFIMA, which explicitly models long memory.

HAR Model (Corsi, 2009). The HAR model captures the multi-scale persistence of realized volatility by aggregating past RV at daily, weekly, and monthly frequencies:

$$RV_{t+h} = \beta_0 + \beta_1 RV_t + \beta_2 RV_{t-5:t} + \beta_3 RV_{t-22:t} + \varepsilon_{t+h}, \tag{1}$$

where $RV_{t-k:t} \equiv k^{-1} \sum_{i=0}^{k-1} RV_{t-i}$ denotes the average realized volatility over the previous k days. We use the original specification, where the weekly and monthly components include overlapping lags (i.e., the weekly component averages days t through $t-4$, not the non-overlapping "rotated" version that separates lags 2–5 from 6–22). This three-component structure approximates long-memory behavior through the actions of heterogeneous market participants operating at different time horizons.

HAR-J Model (Andersen et al., 2007). The HAR-J model augments the baseline HAR with a jump component to separate continuous and discontinuous variation:

$$RV_{t+h} = \beta_0 + \beta_1 RV_t + \beta_2 RV_{t-5:t} + \beta_3 RV_{t-22:t} + \beta_J J_t + \varepsilon_{t+h}, \tag{2}$$

where $J_t = \max(RV_t - BPV_t, 0)$ measures the jump component as the positive part of the difference between realized variance and bipower variation (Barndorff-Nielsen and Shephard, 2004). A negative coefficient on J_t indicates that large jumps reduce, rather than increase, future volatility.

HAR-RS Model (Patton and Sheppard, 2015). The HAR-RS model decomposes realized variance into positive and negative semivariance components to capture asymmetric volatility responses:

$$RV_{t+h} = \beta_0 + \beta_1^+ RS_t^+ + \beta_1^- RS_t^- + \beta_2^+ RS_{t-5:t}^+ + \beta_2^- RS_{t-5:t}^- + \beta_3^+ RS_{t-22:t}^+ + \beta_3^- RS_{t-22:t}^- + \varepsilon_{t+h}, \quad (3)$$

where RS_t^+ and RS_t^- denote good and bad realized semivariance, respectively, and $RS_t^+ + RS_t^- = RV_t$ by construction (Barndorff-Nielsen et al., 2010). The six-regressor structure allows upside and downside risk to follow separate dynamics at each aggregation frequency.

HARQ Model (Bollerslev et al., 2016). The HARQ model accounts for time-varying measurement error in realized volatility by interacting the daily RV regressor with the square root of realized quarticity:

$$RV_{t+h} = \beta_0 + (\beta_1 + \beta_1^Q \sqrt{RQ_t})\, RV_t + \beta_2 RV_{t-5:t} + \beta_3 RV_{t-22:t} + \varepsilon_{t+h}, \quad (4)$$

where RQ_t is the realized quarticity. The interaction term attenuates the daily RV signal when measurement noise is high, as indicated by large values of RQ_t.

Log-HAR Model (Corsi, 2009). The Log-HAR model applies the HAR specification to log-transformed realized volatility:

$$\log RV_{t+h} = \beta_0 + \beta_1 \log RV_t + \beta_2 \log RV_{t-5:t} + \beta_3 \log RV_{t-22:t} + \varepsilon_{t+h}. \quad (5)$$

The logarithmic transformation brings the RV distribution closer to normality, improving ordinary least squares (OLS) efficiency. Point forecasts in levels are recovered via bias-corrected retransformation: $\widehat{RV}_{t+h} = \exp(\hat{y}_{t+h} + \hat{\sigma}^2/2)$, where $\hat{\sigma}^2$ is the estimated residual variance, following the standard log-normal adjustment.

ARFIMA Model (Andersen et al., 2003). The ARFIMA model captures the well-documented long-memory property of realized volatility through fractional differencing:

$$(1-L)^d\,\Phi(L)\,(\log RV_t - \mu) = \Theta(L)\,\varepsilon_t, \tag{6}$$

where $d \in (0, 0.5)$ is the fractional integration parameter, $\Phi(L)$ and $\Theta(L)$ are autoregressive and moving average lag polynomials of orders p and q (with $p, q \leq 2$ selected by the Akaike Information Criterion (AIC)), and L is the lag operator. The parameter d is estimated by maximum likelihood. The fractional differencing filter is truncated at $K = 100$ lags.

All HAR-family models are estimated by OLS with Newey–West heteroskedasticity and autocorrelation consistent standard errors. The ARFIMA model is re-estimated every 22 trading days to balance computational cost against parameter staleness. We do not include the Realized GARCH model of Hansen et al. (2012), which jointly models daily returns and realized measures via maximum likelihood. VOLARE provides realized measures but not the daily close-to-close returns required for the return equation, and sourcing returns from an external provider would introduce alignment issues between the two data sources. This omission is unlikely to affect our main conclusions: Hansen et al. (2012) report that Realized GARCH improves upon standard GARCH but does not consistently dominate HAR-family models in out-of-sample forecasting.

4.2 Time Series Foundation Models

We evaluate eight time series foundation models (TSFMs) in a zero-shot setting, applied directly to realized volatility series without any domain-specific training or fine-tuning. The eight models span six distinct architectures: Chronos-Bolt (small and base checkpoints), Moirai 2.0, Lag-Llama, TimesFM 2.5, Toto, Sundial, and Moirai-MoE. The zero-shot approach is the most practically relevant deployment mode: it requires no labeled financial data, no model retraining, and can be applied immediately to any new asset. Among the

TSFMs we evaluate, Chronos-Bolt, Moirai 2.0, and Sundial are inference-only architectures with no public fine-tuning API. Lag-Llama supports fine-tuning through its GluonTS estimator interface, and the full Chronos-2 model (distinct from the distilled Bolt variant) provides built-in LoRA fine-tuning via its `fit()` method; we leave these adaptation strategies for future work.[23]

Formal framework. A TSFM can be expressed as a parametric mapping from an observed history to a forecast distribution. Let $y_{1:T} = (y_1, \ldots, y_T)$ denote the context window of T past observations (in our case, $T = 512$ daily RV values). A TSFM with pretrained parameters $\hat{\theta}$ produces a forecast of the next H values:

$$\hat{y}_{T+1:T+H} = f_{\hat{\theta}}(y_{1:T}), \tag{7}$$

where $f_{\hat{\theta}}$ maps the input sequence to a predictive distribution over future values.[4] The parameters $\hat{\theta}$ are learned during a pretraining phase on large external corpora of time series spanning weather, energy, retail, transport, and macroeconomic domains. At inference time, the model receives only the context window $y_{1:T}$ and produces forecasts without any task-specific parameter updates—this is the zero-shot setting.

TSFM components. Every TSFM implements the mapping in Equation (7) through three stages: *tokenization*, *encoding*, and *decoding*. First, the raw series $y_{1:T}$ is converted into a sequence of discrete or continuous *tokens*—the model's internal representation of the

[2]We do not evaluate MOMENT (Goswami et al., 2024) because it is pretrained with a masked reconstruction objective rather than an autoregressive forecasting objective. Its forecasting head, a linear projection layer that maps patch embeddings to the forecast horizon, is not pretrained and must be trained on the target series before the model can produce any predictions (Goswami et al., 2024, Section 4.2). This makes it incompatible with our zero-shot evaluation protocol.

[3]Carriero et al. (2024) additionally evaluate Tiny Time Mixers (TTM) (Ekambaram et al., 2024) and TimeGPT (Garza et al., 2024) for macroeconomic forecasting. We exclude TTM due to software version conflicts with our inference environment and TimeGPT because it is a proprietary, closed-source API that does not permit inspection of model weights or training data, precluding reproducibility.

[4]More precisely, $f_{\hat{\theta}}$ outputs a distribution $\hat{p}(y_{T+1:T+H} \mid y_{1:T}; \hat{\theta})$ from which we extract the median as the point forecast. The parameters $\hat{\theta}$ are fixed at their pretrained values throughout our evaluation; no gradient updates occur during inference.

data. Tokenization serves the same role as variable construction in an econometric model: it determines which features of the history the model can "see." The TSFMs we evaluate use different tokenization strategies (quantile binning, patch projection, lag features, flow matching), summarized in Table 2 and described individually below. Second, the token sequence is processed by a *transformer* (Vaswani et al., 2017), a neural network in which each position attends to other positions via learned attention weights, capturing dependencies at arbitrary lags without imposing the specific functional form that HAR prescribes (daily, weekly, monthly components). Third, the transformer's output is passed through a *decoding head* that maps internal representations back to forecasts in the original scale. Decoding heads differ across models: some output quantiles of the predictive distribution directly, others parameterize an explicit probability distribution (e.g., Student-t) from which forecast trajectories are sampled, and one (Sundial) generates forecast trajectories via learned continuous normalizing flows. In all cases, we extract the median as the point forecast, analogous to reporting the conditional median from a quantile regression.

Table 2: Summary of TSFM architectures evaluated. All models are applied zero-shot with a 512-day context window. "Params" refers to the total number of pretrained parameters. The point forecast is the median of each model's predictive distribution.

Model	Architecture	Tokenization	Params	Output	Release
Chronos-Bolt-S	Encoder-only	Quantile binning	48M	Direct quantiles	2025
Chronos-Bolt-B	Encoder-only	Quantile binning	205M	Direct quantiles	2025
Moirai 2.0-S	Decoder-only	Patch projection	14M	9 quantile levels	2025
Lag-Llama	Decoder-only	Lag features	7.5M	Student-t mixture	2024
TimesFM 2.5	Decoder-only	Patch + residual	200M	11 quantile levels	2024
Toto	Decoder-only	Patch projection	162M	Student-t mixture	2024
Sundial	Decoder-only	Flow matching	128M	Generative samples	2025
Moirai-MoE-S	Decoder-only	Patch projection	43M	9 quantile levels	2024

Chronos-Bolt (Ansari et al., 2025). Chronos-Bolt is a distilled variant of the Chronos family. The original Chronos model tokenizes continuous values by mapping them into 4,096 discrete bins via quantile binning, where each observation is replaced by the index of its nearest quantile in a reference distribution, much as one would discretize a continuous

variable into histogram bins. The model then processes these token indices through a T5 encoder-decoder transformer and generates forecasts autoregressively, predicting one token at a time. Chronos-Bolt replaces this two-step architecture with an encoder-only transformer that reads the entire input context and produces all forecast quantiles in a single forward pass, without iterative decoding. This yields a 250× speedup over the original while maintaining competitive accuracy. The decoding head outputs quantile forecasts directly (e.g., the 10th, 50th, and 90th percentiles), bypassing the need to sample from a token distribution. We use both the small (48M parameters) and base (205M parameters) checkpoints[5] and take the median (0.5 quantile) as the point forecast.

Moirai 2.0 (Liu et al., 2025a). Moirai 2.0 builds on the original Moirai architecture. It tokenizes the input series by dividing it into fixed-length *patches* (contiguous blocks of consecutive observations, e.g., 32 days each) and projects each patch into the transformer's embedding space via a learned linear map. This is analogous to constructing non-overlapping moving-window summaries of the series, except that the summary is learned rather than hand-specified. The transformer is decoder-only: it processes patches causally (each patch attends only to itself and earlier patches), preserving the temporal ordering. Moirai 2.0 uses *multi-token prediction*, generating multiple future patches in a single pass rather than one step at a time. The decoding head outputs nine quantile levels (0.05, 0.1, 0.2, 0.3, 0.5, 0.7, 0.8, 0.9, 0.95) per forecast step. We use the small checkpoint[6] and extract the median quantile as the point forecast.

Lag-Llama (Rasul et al., 2024). Lag-Llama is a decoder-only foundation model based on the LLaMA architecture, adapted for univariate probabilistic time series forecasting. Unlike the patch-based models above, Lag-Llama tokenizes each time step individually using a set of *lag features*: at each position t, the input vector consists of the current value and a fixed

[5] `amazon/chronos-bolt-small` and `amazon/chronos-bolt-base` on HuggingFace.
[6] `Salesforce/moirai-2.0-R-small` on HuggingFace.

set of lagged values $(y_{t-1}, y_{t-2}, y_{t-7}, y_{t-14}, \ldots)$, concatenated with time-related covariates. This lag-based design is conceptually closer to an autoregressive econometric specification, since the model explicitly receives lagged values as inputs, though the transformer learns the mapping from lags to forecasts nonparametrically rather than through a linear projection. The decoding head parameterizes a Student-t mixture distribution at each forecast step: it outputs the location, scale, and degrees-of-freedom parameters for each mixture component. Forecasts are generated by sampling trajectories from this distribution and taking the median across samples. We use the pretrained checkpoint (8 layers, 4 attention heads, 36-dimensional embeddings per head), sample 20 trajectories, and report the median as the point forecast.

TimesFM 2.5 (Das et al., 2024). TimesFM 2.5 is Google's decoder-only foundation model for time series forecasting. It uses a patch-based tokenization with residual connections: the input series is divided into fixed-length patches, and each patch embedding includes a residual connection to the raw input values. The model has 200M parameters and outputs 11 quantile levels (0.05, 0.1, ..., 0.9, 0.95) per forecast step. TimesFM was pretrained on a mixture of Google Trends data and public time series corpora. We use the `google/timesfm-2.5-200m-pytorch` checkpoint and extract the median quantile as the point forecast.

Toto (Cohen et al., 2024). Toto is Datadog's decoder-only foundation model, designed for observability and infrastructure time series. It has 162M parameters and uses patch-based tokenization with a learned linear projection. The decoding head parameterizes a Student-t mixture distribution, outputting location, scale, and degrees-of-freedom parameters for each mixture component. Toto was pretrained primarily on observability metrics (CPU usage, request latency, error rates), making it the only TSFM in our study whose training domain is furthest from financial time series. We use the `Datadog/Toto-Open-Base-1.0` checkpoint.

Sundial (Liu et al., 2025b). Sundial is a generative foundation model from THUML (Tsinghua University) that uses flow matching rather than quantile regression for probabilistic forecasting. The model has 128M parameters and generates forecast trajectories by learning a continuous normalizing flow that maps a simple base distribution to the forecast distribution. This generative approach produces full distributional forecasts without discretizing the output space into quantile bins. We use the `thuml/sundial-base-128m` checkpoint, sample 20 trajectories, and report the median as the point forecast.

Moirai-MoE (Liu et al., 2024a). Moirai-MoE extends the Moirai architecture with a sparse Mixture of Experts (MoE) layer that routes each input patch to a subset of specialized expert networks. This allows the model to increase capacity without proportionally increasing inference cost: only a fraction of parameters are active for each input. The small checkpoint has 43M total parameters. Like Moirai 2.0, it uses patch-based tokenization and outputs 9 quantile levels. We use the `Salesforce/moirai-moe-1.0-R-small` checkpoint and extract the median quantile as the point forecast.

Zero-shot protocol. All eight TSFMs receive a rolling context window of 512 daily RV observations as input, with no additional features or covariates. The 512-day context length is the maximum supported by all architectures' positional encodings and is the default recommended setting; it corresponds to approximately two years of trading data, providing enough history to capture the long-memory dynamics of realized volatility. By contrast, the 252-day rolling estimation window for econometric models follows the convention of using one year of daily data, which is standard in the HAR literature. The one-year window balances the need for sufficient data to estimate HAR's four parameters reliably against the risk of including stale observations from a different volatility regime. This reflects each approach's recommended configuration rather than an equal-information comparison; we view this as the practically relevant setup, since a practitioner would deploy each model with its standard settings. We assess sensitivity to the context window length in Section 7.

For multi-step horizons ($h > 1$), the point forecast is the mean of the h-step output vector. No parameters are updated during the out-of-sample evaluation period; all forecasts are purely zero-shot. We extract the median rather than the mean of each model's predictive distribution because the median is the optimal point forecast under MAE loss and is more robust to the right-skewed forecast distributions that arise in volatility applications; for symmetric distributions, the two coincide. We acknowledge that the median is not the optimal point forecast under QLIKE loss (the conditional mean is); however, in practice, the median is the standard output of probabilistic TSFMs and produces more stable forecasts than the sample mean when the predictive distribution is heavy-tailed. Table 3 reports wall-clock timing for all models. Econometric specifications require < 0.01 s per forecast step, run entirely on CPU, and impose no special hardware requirements; they execute on commodity hardware including a standard laptop. TSFM inference was conducted on the Duke Computing Cluster (DCC) using a single NVIDIA RTX A5000 GPU (24 GB VRAM), with per-step times ranging from 0.014 s (Chronos-Bolt-Small) to 0.084 s (Lag-Llama). Even the slowest TSFM produces a forecast in under 0.1 s per asset, making the per-step cost negligible relative to the estimation overhead that econometric models avoid entirely in the zero-shot setting.

4.3 Pretraining Data and Contamination Risk

A natural concern with zero-shot evaluation is whether TSFM performance is inflated because the models' pretraining corpora overlap with the test series. Table 4 summarizes each model's training data and its financial content. No TSFM in our study was trained on realized volatility. VOLARE's realized variance series are second-moment statistics derived from ultra-high-frequency intraday returns at 5-minute sampling (Cipollini et al., 2026); this quantity does not appear in any known training corpus.

Financial data constitutes less than 1% of training observations for all TSFMs we evaluate, limited to daily exchange rates and macroeconomic indicators. This minimal financial

Table 3: Computational cost per forecast step. TSFM inference was run on the Duke Computing Cluster (DCC) using SLURM job scheduling on the `gpu-common` partition, with a single NVIDIA RTX A5000 GPU (24 GB). Econometric models require only CPU and run on commodity hardware (including a standard laptop) in under one minute per asset; their timings here were measured on a DCC node (Intel Xeon Gold 5320, 2.20 GHz). Seconds per step is the wall-clock time to produce one forecast for one asset-day, averaged across the equity sample ($n \approx 2{,}274$ out-of-sample steps). Total time is the approximate wall-clock for one asset across three horizons. ARFIMA is re-estimated every 22 steps. Lag-Llama timing includes 20 sample paths per step.

Model	Parameters	Device	Sec/step (h=1)	Total (1 asset)
Econometric models				
HAR	4	CPU	0.003	21 s
HAR-J	5	CPU	0.003	21 s
HAR-RS	7	CPU	0.004	27 s
HARQ	5	CPU	0.004	24 s
Log-HAR	4	CPU	0.004	30 s
ARFIMA	$\leq$7	CPU	0.006	45 s
Time series foundation models				
Chronos-Bolt-S	48M	GPU	0.014	1.6 min
Chronos-Bolt-B	205M	GPU	0.025	2.8 min
Moirai-2.0-S	14M	GPU	0.019	2.2 min
Lag-Llama	8M	GPU	0.084	9.5 min
TimesFM-2.5	200M	GPU	0.031	3.5 min
Toto	162M	GPU	0.028	3.2 min
Sundial	128M	GPU	0.045	5.1 min
Moirai-MoE-S	43M	GPU	0.022	2.5 min

Table 4: TSFM pretraining corpora and financial data exposure. "Financial Share" reports the fraction of total training observations from financial or economic domains. No model was trained on realized volatility or any intraday-derived statistic.

Model	Corpus Size	Fin. Share	Financial Datasets	RV?
Chronos-Bolt	890K series, 84B obs	<0.3%	`exchange_rate`, FRED-MD	No
Moirai 2.0	27.6B obs (LOTSA)	0.10%	Undisclosed	No
Lag-Llama	8K series	Small	`exchange_rate`	No
TimesFM 2.5	100B+ obs	<1%	Google Trends	No
Toto	Observability data	<0.1%	None (infrastructure metrics)	No
Sundial	Public TS corpora	<1%	Undisclosed	No
Moirai-MoE	27.6B obs (LOTSA)	0.10%	Undisclosed	No

exposure makes data contamination an unlikely explanation for TSFM performance on realized volatility. Instead, the results suggest that TSFM accuracy reflects transfer of general temporal patterns (mean reversion, long memory, regime persistence) rather than memorization of financial dynamics.

4.4 Forecast Evaluation

Walk-forward design. We employ a walk-forward evaluation scheme with a rolling origin. Econometric models use a fixed estimation window of 252 trading days (approximately one year): at each origin, the model is estimated on the most recent 252 observations, and the window slides forward one day at a time, producing daily re-estimated forecasts. Foundation models follow the same walk-forward timing but use a 512-day context window and require no estimation step; only the context window slides forward. Because the TSFM context window is longer, the first TSFM forecast is produced 260 days later than the first econometric forecast; all model comparisons are conducted on the common out-of-sample period where both econometric and TSFM forecasts are available. For the equity sample (2,786 days), this produces approximately 2,148 common out-of-sample forecasts per model–asset combination at $h = 1$; the FX and futures samples (4,240+ days) yield approximately 3,602.

Forecast horizons. We evaluate three forecast horizons: $h = 1$ (one day), $h = 5$ (one week), and $h = 22$ (one month). The forecast target at horizon h is the average realized volatility over the next h days, $h^{-1} \sum_{i=1}^{h} RV_{t+i}$. All models produce direct h-step forecasts. For HAR-family models, this means estimating a separate regression for each horizon, where the dependent variable is the h-day average of future RV; the model does not iterate one-step-ahead predictions forward. ARFIMA similarly produces direct multi-step forecasts from its estimated ARMA representation on log RV. For TSFMs, each model generates a full forecast trajectory of length h in a single inference call; the point forecast is the mean of this h-step output vector. This direct approach avoids the compounding of errors that arises in iterated

multi-step forecasting.

Loss functions and statistical testing. Our primary loss function is the quasi-likelihood (QLIKE) loss of Patton (2011):

$$L_{\text{QLIKE}}(\hat{\sigma}_t^2, RV_t) = \frac{RV_t}{\hat{\sigma}_t^2} - \log \frac{RV_t}{\hat{\sigma}_t^2} - 1, \tag{8}$$

where $\hat{\sigma}_t^2$ is the forecast and RV_t is the realized value. QLIKE preserves the ranking of forecasts regardless of the choice of volatility proxy, making it the preferred metric when the true conditional variance is unobserved. We also report mean squared error (MSE), mean absolute error (MAE), and the out-of-sample R^2 defined as $R^2_{\text{OOS}} = 1 - \sum_t (RV_t - \hat{f}_t)^2 / \sum_t (RV_t - \bar{RV})^2$, where $\bar{RV}$ is the historical mean of realized variance over the estimation window. Unlike the in-sample R^2, R^2_{OOS} can take negative values, indicating that the model's forecasts are less accurate than a simple historical mean forecast. All forecasts are floored at 10^{-10} to prevent numerical instability in the QLIKE calculation when models produce near-zero predictions.

Pairwise forecast comparisons use the DM test with Newey–West variance estimation, applied to QLIKE loss differentials. To address the multiple comparison problem inherent in evaluating fourteen models, we compute the MCS, which identifies the subset of models whose forecasting ability cannot be statistically distinguished from the best model at a given significance level. We use the $T_{\max}$ test statistic with a block bootstrap (block length = 22 days, $B = 10{,}000$ replications) and significance level $\alpha = 0.10$.

We assess forecast efficiency using the Mincer–Zarnowitz (MZ) regression (Mincer and Zarnowitz, 1969):

$$RV_t = \alpha + \beta \, \hat{f}_t + \varepsilon_t, \tag{9}$$

where $\hat{f}_t$ is the model's forecast. Under forecast optimality, $\alpha = 0$ and $\beta = 1$, meaning the forecast is unbiased and captures the correct scale of variation. We test the joint null

$H_0 \colon \alpha = 0, \beta = 1$ using a Wald test with Newey–West standard errors and report cross-asset averages and rejection rates at the 5% level.

To examine whether relative forecast performance is stable over time, we apply the Giacomini and Rossi (2010) fluctuation test (GR). For each model paired against the HAR benchmark, we compute a rolling DM statistic over a window of size $m = \lfloor 0.3 \times T \rfloor$, producing a time path of relative QLIKE performance. The test statistic is $\sup_t |S_t|$, compared against critical values from the distribution of the supremum of a standardized Brownian bridge. Rejection of the null indicates that the two models' relative performance changes significantly over time.

5 Empirical Results

This section presents the univariate realized variance forecasting results. We first examine aggregate forecast accuracy across loss functions and horizons, then analyze cross-asset and cross-market heterogeneity.

5.1 Aggregate Forecast Accuracy

This subsection reports the overall forecasting performance averaged across assets within each asset class, without examining individual asset-level variation. The goal is to establish the broad ranking of models by loss function and horizon before turning to cross-asset heterogeneity and statistical testing in subsequent subsections.

Table 5 reports the cross-sectional average of per-asset loss functions across 40 equities. Figure 1 illustrates the forecasts of three representative models against realized values for four assets spanning equities and FX. The results reveal a systematic divergence between QLIKE and the other metrics.

Table 5: Forecast accuracy for 40 U.S. equities (VOLARE). Values are cross-sectional averages of per-asset loss functions. Bold indicates the best value in each column within each panel. † denotes inflated QLIKE due to near-zero forecasts or systematic forecast bias.

Model	MSE ($\times 10^{-6}$)	MAE ($\times 10^{-4}$)	QLIKE	R^2_{OOS}
Panel: $h = 1$ (1 day)				
HAR	0.243	1.42	1.38†	0.339
HAR-J	0.241	1.41	1.11†	0.353
HAR-RS	0.264	1.51	3.63†	0.274
HARQ	0.296	1.49	16.04†	0.116
Log-HAR	0.255	1.31	0.219	0.246
ARFIMA	0.300	1.63	0.322	0.131
Chronos-Bolt-S	0.216	1.18	0.229	0.408
Chronos-Bolt-B	0.213	1.17	0.230	0.420
Moirai-2.0-S	0.210	1.19	0.226	0.426
Lag-Llama	0.291	1.65	0.402	0.127
TimesFM-2.5	**0.208**	**1.16**	0.222	**0.442**
Toto	0.301	2.41	34.59†	-0.011
Sundial	0.210	1.21	**0.195**	0.431
Moirai-MoE-S	0.214	1.23	0.211	0.419
Panel: $h = 5$ (5 days)				
HAR	0.161	1.39	2.70†	0.217
HAR-J	0.158	1.38	2.47†	0.238
HAR-RS	0.183	1.49	6.83†	0.110
HARQ	0.190	1.42	15.89†	0.033
Log-HAR	0.161	1.26	0.229	0.150
ARFIMA	0.175	1.44	0.251	0.143
Chronos-Bolt-S	0.128	1.09	0.213	0.354
Chronos-Bolt-B	0.128	1.09	0.216	0.346
Moirai-2.0-S	0.122	1.07	0.188	0.384
Lag-Llama	0.181	1.43	0.305	0.044
TimesFM-2.5	0.122	**1.06**	0.207	0.385
Toto	0.169	1.67	16.14†	0.013
Sundial	**0.120**	1.09	**0.161**	**0.394**
Moirai-MoE-S	0.122	1.08	0.167	0.372
Panel: $h = 22$ (22 days)				
HAR	0.116	1.46	5.92†	-0.017
HAR-J	0.115	1.45	5.79†	-0.003
HAR-RS	0.128	1.55	9.28†	-0.129
HARQ	0.130	1.47	12.02†	-0.168
Log-HAR	0.113	1.34	0.345	-0.025
ARFIMA	0.123	1.34	0.280	-0.060
Chronos-Bolt-S	0.101	1.13	0.322	0.085
Chronos-Bolt-B	0.101	1.14	0.325	0.077
Moirai-2.0-S	0.098	**1.07**	0.257	0.109
Lag-Llama	0.109	1.25	0.283	0.009
TimesFM-2.5	0.099	1.11	0.326	0.098
Toto	0.281	1.61	6.44†	-2.300
Sundial	**0.092**	1.08	**0.218**	**0.148**
Moirai-MoE-S	0.139	1.14	0.233	-0.417

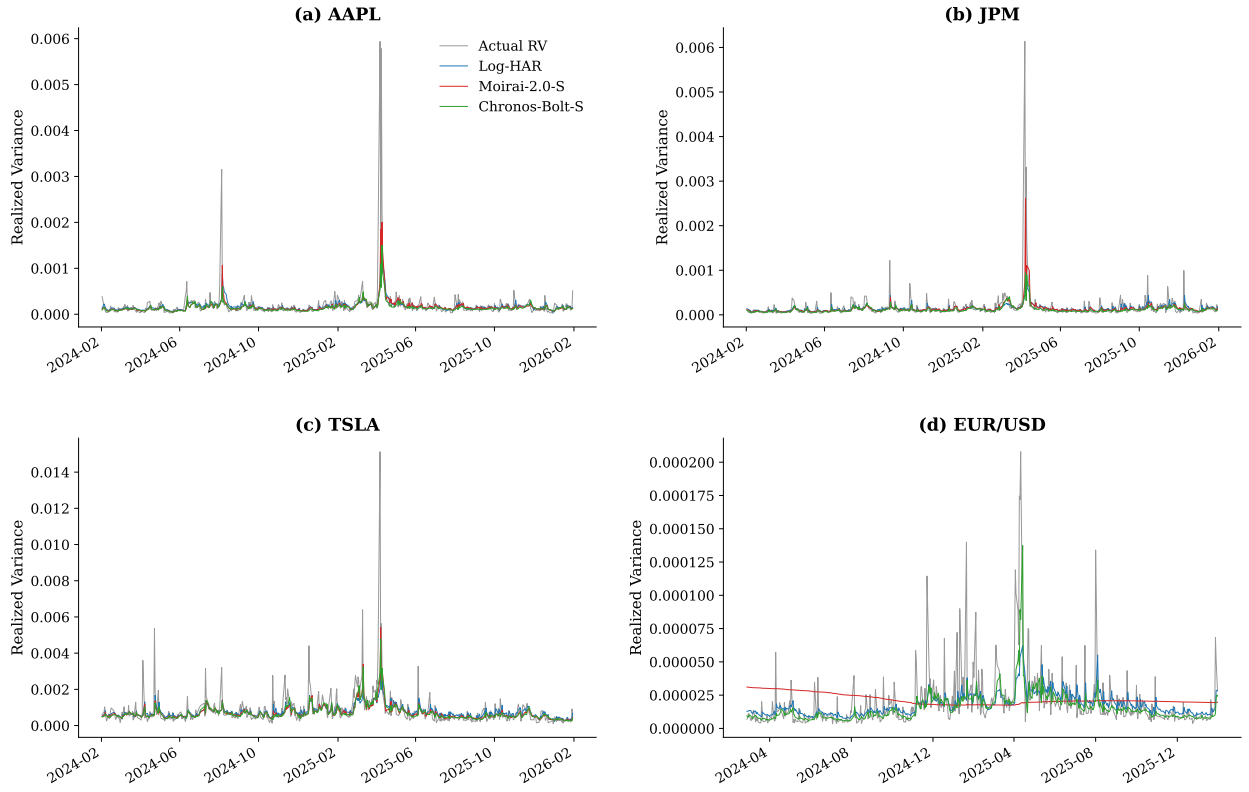

Figure 1: Forecast vs. actual realized variance at $h = 1$ for four representative assets. The last 500 out-of-sample observations are shown. Gray: actual RV; blue: Log-HAR; red: Moirai 2.0; green: Chronos-Bolt-Small.

At the one-day horizon ($h = 1$), Sundial achieves the lowest average QLIKE (0.195), followed by Moirai-MoE-S (0.211), Log-HAR (0.219), and TimesFM 2.5 (0.222). The gap between Sundial and the best econometric model (Log-HAR) is 11% in relative terms, a substantial margin for a zero-shot model with no financial training data. On MSE, MAE, and R^2_{OOS}, TSFMs again dominate: TimesFM 2.5 achieves the lowest MSE (0.208×10^{-6}) and the highest R^2_{OOS} (0.442), while Sundial (0.431) and Moirai 2.0 (0.426) follow closely. The out-of-sample R^2 gap between the best TSFMs and the best econometric model (HAR-J, 0.353) is 7–9 percentage points at $h = 1$. However, not all TSFMs perform well: Toto produces severely miscalibrated forecasts (QLIKE = 34.59, $R^2_{\text{OOS}} = -0.011$), driven primarily by scale misalignment rather than absence of signal (Section 7), and Lag-Llama underperforms all

econometric benchmarks except ARFIMA on QLIKE (0.402).

At the five-day horizon ($h = 5$), the TSFM advantage becomes more pronounced. Sundial achieves the best QLIKE (0.161), MSE (0.120×10^{-6}), and R^2_{OOS} (0.394), with Moirai-MoE-S close behind (QLIKE = 0.167). TimesFM 2.5 achieves the lowest MAE (1.06×10^{-4}) and ties Moirai 2.0 for second-lowest MSE (0.122×10^{-6}). The gap over Log-HAR in QLIKE terms is now 30% for Sundial, compared to 11% at $h = 1$. The econometric models that do best at $h = 1$ (HAR, HAR-J) deteriorate more rapidly at longer horizons, while the top TSFMs maintain relatively stable accuracy.

At the monthly horizon ($h = 22$), Sundial again dominates: QLIKE = 0.218, MSE = 0.092×10^{-6}, and $R^2_{\text{OOS}} = 0.148$. Moirai-MoE-S achieves the second-lowest QLIKE (0.233), followed by Moirai 2.0 (0.257). The econometric benchmarks now show near-zero or negative R^2_{OOS}, indicating that they barely outperform (or underperform) the historical mean at this horizon. Log-HAR achieves $R^2_{\text{OOS}} = -0.025$, while Sundial maintains a positive R^2_{OOS} of 0.148. This horizon-dependent pattern is consistent with the hypothesis that TSFMs' large context windows (512 days) capture long-range dependencies more effectively than the HAR's fixed three-component structure.

Table 6: Forecast accuracy for 40 U.S. equities (VOLARE), cross-sectional medians. Median aggregation is robust to outlier assets with degenerate forecasts. Bold indicates the best value in each column within each panel.

Model	MSE ($\times 10^{-6}$)	MAE ($\times 10^{-4}$)	QLIKE	R^2_{OOS}
Panel: $h = 1$ (1 day)				
HAR	0.116	1.13	0.219	0.367
HAR-J	0.113	1.14	0.445	0.365
HAR-RS	0.124	1.19	$1.05^{\dagger}$	0.321
HARQ	0.173	1.24	$10.30^{\dagger}$	0.066
Log-HAR	0.139	1.12	0.208	0.237
ARFIMA	0.148	1.37	0.299	0.163
Chronos-Bolt-S	0.096	0.97	0.212	0.436
Chronos-Bolt-B	0.094	0.95	0.213	0.459
Moirai-2.0-S	0.092	0.98	0.209	0.451
Lag-Llama	0.156	1.38	0.384	0.134
TimesFM-2.5	**0.090**	**0.95**	0.203	0.471
Toto	0.175	2.19	$36.25^{\dagger}$	0.029
Sundial	0.094	1.00	**0.181**	**0.473**
Moirai-MoE-S	0.098	1.01	0.195	0.460
Panel: $h = 5$ (5 days)				
HAR	0.096	1.12	0.269	0.265
HAR-J	0.094	1.10	0.381	0.284
HAR-RS	0.109	1.19	$3.39^{\dagger}$	0.191
HARQ	0.126	1.17	$11.81^{\dagger}$	0.017
Log-HAR	0.111	1.08	0.205	0.139
ARFIMA	0.099	1.19	0.234	0.192
Chronos-Bolt-S	0.077	0.90	0.196	0.366
Chronos-Bolt-B	0.080	0.91	0.196	0.359
Moirai-2.0-S	0.073	**0.86**	0.172	0.383
Lag-Llama	0.128	1.20	0.295	0.039
TimesFM-2.5	0.073	0.88	0.189	0.392
Toto	0.130	1.48	$17.72^{\dagger}$	0.061
Sundial	0.078	0.91	**0.149**	**0.400**
Moirai-MoE-S	**0.070**	0.90	0.149	0.380
Panel: $h = 22$ (22 days)				
HAR	0.078	1.20	$3.24^{\dagger}$	0.013
HAR-J	0.076	1.20	$3.33^{\dagger}$	0.029
HAR-RS	0.082	1.27	$5.23^{\dagger}$	-0.087
HARQ	0.089	1.25	$8.65^{\dagger}$	-0.116
Log-HAR	0.081	1.11	0.296	-0.029
ARFIMA	0.073	1.10	0.274	0.073
Chronos-Bolt-S	0.071	0.94	0.319	0.085
Chronos-Bolt-B	0.072	0.97	0.326	0.083
Moirai-2.0-S	**0.069**	**0.89**	0.249	0.110
Lag-Llama	0.079	1.03	0.267	0.002
TimesFM-2.5	0.071	0.93	0.320	0.105
Toto	0.254	1.51	$7.06^{\dagger}$	-2.030
Sundial	0.070	0.93	**0.217**	**0.143**
Moirai-MoE-S	0.110	0.99	0.232	-0.340

Table 6 reports cross-sectional medians, which eliminate the influence of outlier assets where levels-based HAR variants produce near-zero forecasts. Under median aggregation, the qualitative model ranking is preserved: Sundial achieves the best median QLIKE at $h = 1$ (0.181), followed by Moirai-MoE-S (0.195) and TimesFM 2.5 (0.203). Sundial and Moirai-MoE-S are tied for the lowest median QLIKE at $h = 5$ (both 0.149), while Sundial dominates at $h = 22$ (0.217). The gap between TSFMs and econometric models narrows under medians—the median QLIKE difference between Log-HAR and Sundial at $h = 5$ shrinks to 0.056 (vs. 0.068 in means)—confirming that the main findings are not driven by a few outlier assets. The median results also show that the QLIKE inflation of HAR-RS and HARQ is concentrated in a subset of assets: median QLIKE for HARQ drops from 16.04 (mean) to 10.30 (median) at $h = 1$, indicating that the inflation is widespread but reduced in magnitude.

Levels-based HAR extensions are unstable in this evaluation. Unconstrained OLS estimation of HAR variants can produce negative or near-zero forecasts. HAR-RS and HARQ produce QLIKE values of 3.63 and 16.04 at $h = 1$, respectively, rising to 9.28 and 12.02 at $h = 22$. These inflated values arise because the OLS-estimated models occasionally produce near-zero forecasts for RV_{t+h}, which are floored at 10^{-10} for QLIKE evaluation. Across the full sample, no forecast is literally negative, but the floor binds for a small fraction of observations. Since QLIKE $= RV/\hat{RV} - \log(RV/\hat{RV}) - 1$, a forecast near zero generates an extremely large loss whenever the realized value is far from zero, and a handful of such observations can dominate the cross-sectional average. Across the equity sample, the 10^{-10} floor binds for fewer than 0.1% of TSFM forecast–date pairs but for 2–5% of HAR-RS and HARQ pairs, explaining their inflated QLIKE. The median results in Table 6 confirm that the qualitative ranking of models is unchanged. Log-HAR and ARFIMA avoid this problem entirely by operating in log space, where forecasts are positive by construction. TSFMs also produce strictly positive forecasts, since their training distributions are bounded below. This practical advantage may be as important as any differences in statistical accuracy: a

model that occasionally produces near-zero forecasts is difficult to deploy in production risk systems.

Figure 2 provides a distributional view of these results, plotting QLIKE ratios (model / HAR) across all 50 assets for selected models. Values below one indicate the model outperforms HAR. At $h = 1$, Sundial has the lowest median ratio and the tightest interquartile range, while Lag-Llama's median ratio exceeds 1.7. At $h = 5$, Sundial and Moirai-MoE-S pull ahead of the field, with the vast majority of assets below one. By $h = 22$, all well-calibrated models improve substantially relative to HAR, with Sundial maintaining the lowest median ratio. The box plots also reveal heterogeneity within the TSFM class: Toto's ratio consistently exceeds HAR across all horizons, confirming its severe miscalibration, while Sundial and Moirai-MoE-S show the tightest distributions, indicating reliable cross-asset performance.

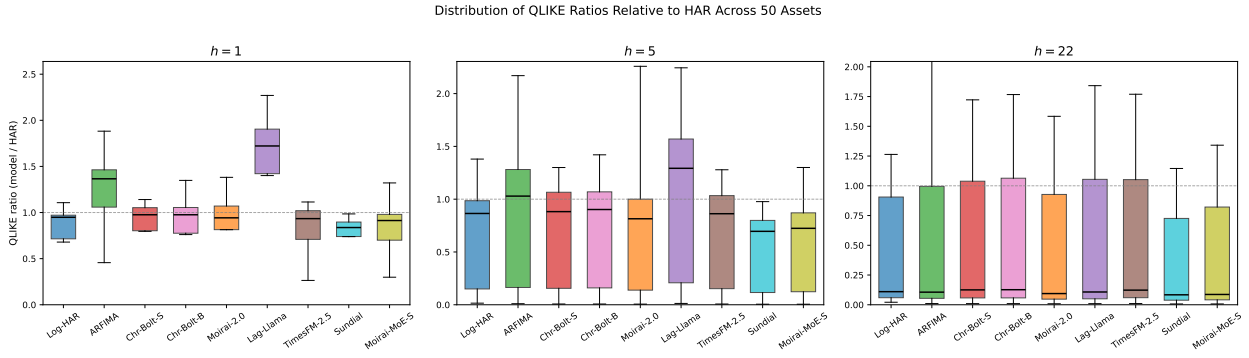

Figure 2: Distribution of QLIKE ratios (model / HAR) across 50 assets. Values below one indicate the model outperforms HAR. Boxes span the interquartile range, whiskers extend to 1.5× IQR, and outliers are suppressed for clarity. The dashed line marks parity with HAR.

5.2 Cross-Asset and Cross-Market Heterogeneity

The equity results above establish a clear ranking, but it is not obvious that these patterns generalize to asset classes with different volatility dynamics. Tables 7 and 8 extend the analysis to foreign exchange and futures markets.

Table 7: Forecast accuracy for 5 FX pairs (VOLARE). Values are averages across the five currency pairs. Format as Table 5.

Model	MSE ($\times 10^{-8}$)	MAE ($\times 10^{-4}$)	QLIKE	R^2_{OOS}
Panel: $h = 1$ (1 day)				
HAR	0.174	0.14	0.177	0.241
HAR-J	0.170	0.14	0.196	0.258
HAR-RS	0.192	0.15	0.249	0.180
HARQ	0.196	0.14	0.628	0.167
Log-HAR	0.174	0.13	0.163	0.247
ARFIMA	0.207	0.16	0.242	0.107
Chronos-Bolt-S	0.168	0.12	0.174	0.277
Chronos-Bolt-B	0.167	0.12	0.176	0.279
Moirai-2.0-S	0.222	0.19	0.329	-0.008
Lag-Llama	0.207	0.17	0.305	0.080
TimesFM-2.5	0.166	**0.12**	0.171	0.284
Toto	3.747	1.21	12.65†	-33.916
Sundial	**0.162**	0.12	**0.153**	**0.301**
Moirai-MoE-S	0.178	0.14	0.199	0.212
Panel: $h = 5$ (5 days)				
HAR	0.082	0.12	0.143	0.284
HAR-J	0.079	0.12	0.149	0.305
HAR-RS	0.098	0.13	0.240	0.173
HARQ	0.085	0.12	0.399	0.259
Log-HAR	0.080	0.11	0.120	0.306
ARFIMA	0.098	0.13	0.178	0.147
Chronos-Bolt-S	0.075	0.10	0.128	0.353
Chronos-Bolt-B	0.075	0.10	0.129	0.353
Moirai-2.0-S	0.110	0.17	0.250	-0.025
Lag-Llama	0.098	0.14	0.194	0.117
TimesFM-2.5	0.074	**0.10**	0.126	0.361
Toto	0.881	0.62	10.97†	-11.826
Sundial	**0.068**	0.10	**0.098**	**0.413**
Moirai-MoE-S	0.076	0.11	0.139	0.300
Panel: $h = 22$ (22 days)				
HAR	0.052	0.13	0.186	0.171
HAR-J	0.052	0.12	0.197	0.177
HAR-RS	0.065	0.13	0.223	0.004
HARQ	0.053	0.12	0.367	0.155
Log-HAR	0.049	0.12	0.147	0.205
ARFIMA	0.060	0.12	0.185	0.033
Chronos-Bolt-S	0.047	0.10	0.161	0.250
Chronos-Bolt-B	0.047	0.10	0.163	0.248
Moirai-2.0-S	0.064	0.16	0.212	-0.093
Lag-Llama	0.053	0.13	0.164	0.119
TimesFM-2.5	0.046	**0.10**	0.165	0.253
Toto	0.722	0.37	9.23†	-10.356
Sundial	**0.042**	0.10	**0.112**	**0.327**
Moirai-MoE-S	0.062	0.11	0.145	-0.063

Table 8: Forecast accuracy for 5 futures contracts (VOLARE). Values are averages across the five contracts. Format as Table 5.

Model	MSE ($\times 10^{-6}$)	MAE ($\times 10^{-4}$)	QLIKE	R^2_{OOS}
Panel: $h = 1$ (1 day)				
HAR	14.818	2.72	0.225	0.244
HAR-J	14.958	2.65	0.707	0.258
HAR-RS	15.503	2.81	$5.82^{\dagger}$	0.180
HARQ	15.156	2.65	$22.34^{\dagger}$	0.089
Log-HAR	14.143	2.09	0.193	0.225
ARFIMA	16.437	2.95	0.346	0.009
Chronos-Bolt-S	14.335	1.99	0.213	0.237
Chronos-Bolt-B	14.251	1.95	0.221	0.246
Moirai-2.0-S	14.509	1.99	0.289	0.213
Lag-Llama	14.860	2.77	$1.12^{\dagger}$	0.074
TimesFM-2.5	**14.084**	**1.92**	0.204	**0.260**
Toto	15.080	3.23	$24.90^{\dagger}$	-0.544
Sundial	14.246	2.05	**0.189**	0.231
Moirai-MoE-S	21.772	2.19	0.211	0.151
Panel: $h = 5$ (5 days)				
HAR	4.574	2.56	0.434	0.253
HAR-J	4.918	2.48	0.839	0.245
HAR-RS	4.615	2.60	$4.07^{\dagger}$	0.161
HARQ	4.792	2.49	$22.10^{\dagger}$	0.181
Log-HAR	4.442	1.98	0.165	0.226
ARFIMA	6.262	2.70	0.269	-0.040
Chronos-Bolt-S	4.352	1.80	0.185	0.273
Chronos-Bolt-B	4.384	1.77	0.196	0.275
Moirai-2.0-S	4.523	1.80	0.207	0.246
Lag-Llama	4.859	2.48	0.333	0.072
TimesFM-2.5	4.262	**1.74**	0.175	0.306
Toto	4.202	2.35	$12.92^{\dagger}$	-0.201
Sundial	**3.977**	1.79	**0.139**	0.299
Moirai-MoE-S	4.501	1.76	0.155	**0.316**
Panel: $h = 22$ (22 days)				
HAR	2.169	2.77	$1.08^{\dagger}$	0.078
HAR-J	2.234	2.73	$1.23^{\dagger}$	0.071
HAR-RS	2.201	2.89	$4.01^{\dagger}$	-0.034
HARQ	2.126	2.66	$17.88^{\dagger}$	-0.051
Log-HAR	2.001	2.16	0.280	0.130
ARFIMA	3.681	2.65	0.286	-0.296
Chronos-Bolt-S	1.873	1.92	0.282	0.168
Chronos-Bolt-B	1.886	1.87	0.286	0.167
Moirai-2.0-S	1.855	1.83	0.238	**0.201**
Lag-Llama	2.062	2.29	0.336	0.082
TimesFM-2.5	1.816	1.85	0.275	0.200
Toto	1.598	2.19	$6.37^{\dagger}$	-1.218
Sundial	**1.595**	1.84	**0.184**	0.097
Moirai-MoE-S	1.625	**1.78**	0.206	0.169

Foreign exchange. The FX sample covers five major currency pairs with lower volatility levels ($\bar{RV} \approx 3 \times 10^{-5}$) and moderate persistence ($\bar{\rho}_1 = 0.52$). For FX, Sundial dominates across all horizons and metrics. At $h = 1$, Sundial achieves the lowest QLIKE (0.153) and MSE (0.162×10^{-8}), followed by Log-HAR (0.163) and TimesFM 2.5 (0.171). Moirai 2.0 underperforms substantially (QLIKE = 0.329, $R^2_{\text{OOS}} = -0.008$). At $h = 5$, Sundial extends its lead (QLIKE = 0.098 vs. 0.120 for Log-HAR), with TimesFM 2.5 (0.126) and Chronos-Bolt (0.128–0.129) close behind. At $h = 22$, Sundial again leads on QLIKE (0.112) and R^2_{OOS} (0.327), while Moirai-MoE-S (0.145) outperforms Log-HAR (0.147). Moirai 2.0's poor FX performance ($R^2_{\text{OOS}} < 0$ at all horizons) is not driven by a single currency pair: per-asset QLIKE analysis confirms that Moirai underperforms all competing models on every FX pair at every horizon. This systematic weakness stands in contrast to its equity strength and suggests that Moirai's training data or architecture is less suited to the smoother, lower-amplitude dynamics of currency volatility.

Futures. The futures sample displays the widest cross-asset heterogeneity in volatility dynamics. The E-mini S&P 500 (ES) has first-order autocorrelation of 0.779, while Gold (GC) shows near-zero autocorrelation ($\rho_1 = -0.001$). At $h = 1$, Sundial achieves the lowest QLIKE (0.189), followed by Log-HAR (0.193) and TimesFM 2.5 (0.204). Moirai 2.0 ranks behind (0.289). At $h = 5$, Sundial leads on QLIKE (0.139), followed by Moirai-MoE-S (0.155) and Log-HAR (0.165); Moirai-MoE-S achieves the highest R^2_{OOS} (0.316). At the monthly horizon, Sundial retains the QLIKE lead (0.184), followed by Moirai-MoE-S (0.206), Moirai 2.0 (0.238), and TimesFM 2.5 (0.275), while levels-based HAR variants produce inflated QLIKE (HAR: 1.08, HARQ: 17.88).

Cross-asset patterns. Three patterns emerge from the cross-asset comparison. First, Sundial is the top-performing model across all three asset classes and all horizons on QLIKE, a level of consistency not achieved by any other model in the evaluation. Second, Log-HAR remains the best or near-best econometric specification across all asset classes, confirming

the practical value of the log transformation for ensuring forecast positivity and aligning with QLIKE's loss structure. Third, within the TSFM class, performance dispersion is wide: Sundial, Moirai-MoE-S, TimesFM 2.5, and Moirai 2.0 form a competitive tier, while Toto produces severely miscalibrated forecasts across all asset classes and Lag-Llama underperforms at short horizons. These results highlight the importance of both multi-market and multi-model evaluations: a model that appears dominant on one asset class may not generalize, and the choice of TSFM architecture matters at least as much as the choice between TSFMs and econometric models.

5.3 Drivers of TSFM Forecast Performance

The cross-asset results above show that TSFM rankings vary across asset classes, raising the question of what asset-level characteristics drive relative forecast performance. Carriero et al. (2024) find that TSFMs struggle with highly persistent macroeconomic series (first-order partial autocorrelation above 0.7). We test whether a similar pattern holds for realized volatility by examining the relationship between each asset's first-order autocorrelation ρ_1 and the TSFM's QLIKE ratio relative to HAR.

Figure 3 plots QLIKE ratios (model / HAR) at $h = 1$ against ρ_1 for 50 assets, separately for each TSFM. Unlike the macroeconomic setting, we find no statistically significant relationship between persistence and TSFM relative performance: the rank correlation ranges from $r = 0.06$ ($p = 0.68$) for Chronos-Bolt to $r = -0.17$ ($p = 0.23$) for Lag-Llama, with the four new models showing similarly insignificant correlations. Sundial beats HAR for 100% of assets in both high- and low-persistence groups, confirming its broad dominance regardless of persistence. Moirai-MoE-S beats HAR for 84% of high-persistence and 68% of low-persistence assets, while TimesFM 2.5 shows a modest persistence effect (76% vs. 56%).

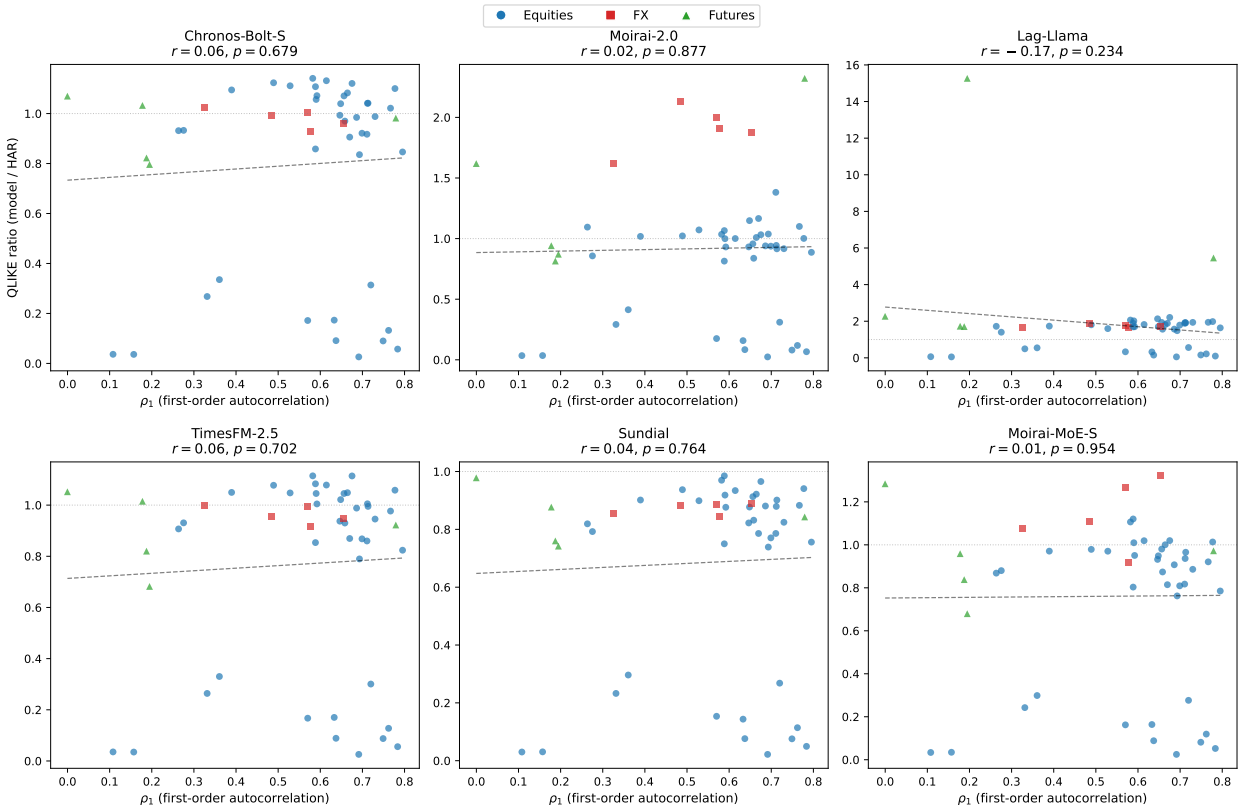

Figure 3: QLIKE ratio (model / HAR) at $h = 1$ vs. first-order autocorrelation ρ_1 for 50 assets. Values below one indicate the TSFM outperforms HAR. Each point is one asset, colored by class (blue: equities, red: FX, green: futures). Dashed line: OLS trend; r and p are the Pearson correlation and its p-value.

This null result is informative. Realized volatility, unlike many macroeconomic series, is stationary and mean-reverting by construction. Even the least persistent assets in our sample (Gold, $\rho_1 \approx 0$) exhibit the positive, right-skewed distribution that TSFMs encounter frequently in their pretraining corpora. The variation in TSFM performance across assets is better explained by architecture than by persistence: Sundial dominates across all asset classes regardless of individual ρ_1, while other TSFMs show asset-class-dependent strengths. This suggests that the relevant driver is the match between each TSFM's implicit prior (shaped by its training data composition and output mechanism) and the distributional characteristics of realized volatility, rather than the degree of serial dependence.

6 Statistical Significance and Model Confidence Sets

The rankings in Section 5 show large differences across models, but mean loss comparisons can be misleading when distributions are skewed and sample sizes vary across assets. We now apply four formal statistical tests: MCS to identify the subset of models that cannot be distinguished from the best, pairwise DM tests to quantify directional win rates, MZ regressions to assess forecast efficiency, and GR fluctuation tests to detect time variation in relative performance.

Table 9 reports MCS inclusion rates for 40 equities, Figure 4 visualizes the inclusion patterns, and Table 10 summarizes pairwise DM test results across all 50 assets.

Table 9: Model Confidence Set inclusion rates for 40 U.S. equities (VOLARE). Each cell reports the percentage of stocks for which the model is included in the MCS at the 10% significance level (QLIKE loss, $T_{\max}$ statistic, block bootstrap with $B = 10{,}000$).

Model	$h = 1$ (%)	$h = 5$ (%)	$h = 22$ (%)
HAR	5.0	10.3	15.0
HAR-J	0.0	7.7	15.0
HAR-RS	0.0	0.0	7.5
HARQ	0.0	0.0	0.0
Log-HAR	35.0	17.9	57.5
ARFIMA	0.0	5.1	67.5
Chronos-Bolt-S	0.0	0.0	15.0
Chronos-Bolt-B	0.0	0.0	12.5
Moirai-2.0-S	7.5	25.6	67.5
Lag-Llama	0.0	0.0	67.5
TimesFM-2.5	0.0	0.0	15.0
Toto	0.0	0.0	2.5
Sundial	100.0	100.0	97.5
Moirai-MoE-S	22.5	71.8	80.0

Table 10: Diebold–Mariano test: pairwise win rates. Each cell reports the percentage of pairwise comparisons (across 50 assets $\times$ 13 opponents) in which the row model achieves significantly lower QLIKE at the 5% level.

Model	$h = 1$	$h = 5$	$h = 22$
HAR	35.4%	24.3%	9.8%
HAR-J	24.9%	22.3%	9.5%
HAR-RS	17.5%	10.3%	5.5%
HARQ	5.5%	4.5%	2.5%
Log-HAR	57.1%	38.6%	23.1%
ARFIMA	26.6%	18.0%	16.6%
Chronos-Bolt-S	40.9%	29.1%	14.0%
Chronos-Bolt-B	40.5%	26.5%	12.6%
Moirai-2.0-S	47.5%	45.7%	27.5%
Lag-Llama	18.2%	16.5%	18.5%
TimesFM-2.5	53.4%	35.5%	12.6%
Toto	0.3%	0.6%	1.7%
Sundial	87.8%	78.8%	51.1%
Moirai-MoE-S	60.3%	66.0%	37.1%

Each cell reports the percentage of pairwise DM tests (across 50 assets $\times$ 13 opponents = 650 tests) in which the row model has significantly lower QLIKE at the 5% level.

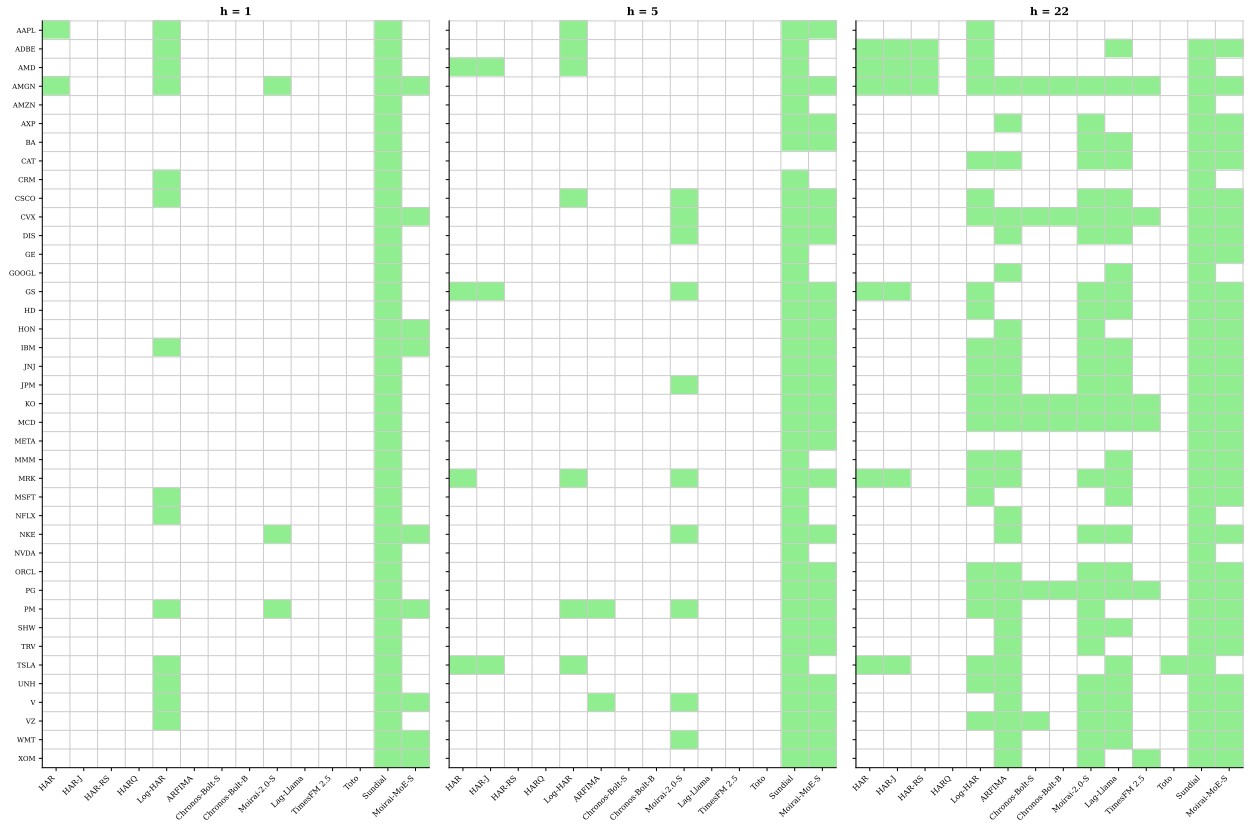

Figure 4: Model Confidence Set inclusion for 40 U.S. equities. Green cells indicate that the model is included in the MCS at the 10% significance level (QLIKE loss). Left: $h = 1$; center: $h = 5$; right: $h = 22$.

MCS results for equities. At $h = 1$, Sundial enters the MCS for 100% of assets, confirming its status as the single most reliable specification at the daily horizon. Log-HAR enters for 35%, Moirai-MoE-S for 22.5%, and Moirai 2.0 for 7.5%. Nine of fourteen models enter for 0%, including all levels-based HAR variants, Chronos-Bolt, Lag-Llama, TimesFM 2.5, and Toto. No other model enters the MCS for more than a third of assets at this horizon.

At $h = 5$, the picture remains TSFM-dominated. Sundial again enters for 100% of assets, followed by Moirai-MoE-S (71.8%), Moirai 2.0 (25.6%), and Log-HAR (17.9%). Chronos-Bolt, TimesFM 2.5, Lag-Llama, and Toto are excluded from all MCS sets.

At $h = 22$, the ranking broadens. Sundial enters for 97.5%, Moirai-MoE-S for 80%,

Moirai 2.0 for 67.5%, Lag-Llama for 67.5%, ARFIMA for 67.5%, and Log-HAR for 57.5%. Lag-Llama's sharp improvement from 0% at $h = 1$ to 67.5% at $h = 22$ suggests that its lag-based tokenization captures long-range dependencies more effectively than short-term dynamics. The high ARFIMA inclusion rate at $h = 22$ is consistent with its long-memory parameterization: at the monthly horizon, fractional differencing captures slow mean reversion that the HAR's three fixed lags approximate less precisely.

DM pairwise tests. The DM test is computed pairwise: for each of the 14 models, we test it against each of the remaining 13 models on each of the 50 assets, yielding $14 \times 13 \times 50 =$ 9,100 individual tests. Table 10 reports the fraction of pairwise DM tests in which each model achieves significantly lower QLIKE at the 5% level, aggregated across 50 assets and 13 opponents. At $h = 1$, Sundial wins the highest fraction of pairwise comparisons (87.8%), followed by Moirai-MoE-S (60.3%), Log-HAR (57.1%), and TimesFM 2.5 (53.4%). Sundial's win rate is 28 percentage points above the next-best model, indicating that its QLIKE advantage is both large and statistically significant across the vast majority of asset–model pairs. At $h = 5$ and $h = 22$, Sundial again achieves the highest win rate (78.8% and 51.1%), followed by Moirai-MoE-S (66.0% and 37.1%). Toto's win rate never exceeds 1.7%, confirming its persistent miscalibration. Lag-Llama's win rate is 18.2% at $h = 1$ and 18.5% at $h = 22$; the modest DM win rate at $h = 22$, despite 67.5% MCS inclusion, reflects the fact that forecast differences narrow at longer horizons, making it harder to reject equal predictive ability in pairwise tests. Figure 5 plots the cumulative QLIKE differential between Log-HAR and Moirai 2.0 at $h = 1$, showing that Moirai's advantage accumulates steadily after the COVID-19 onset.

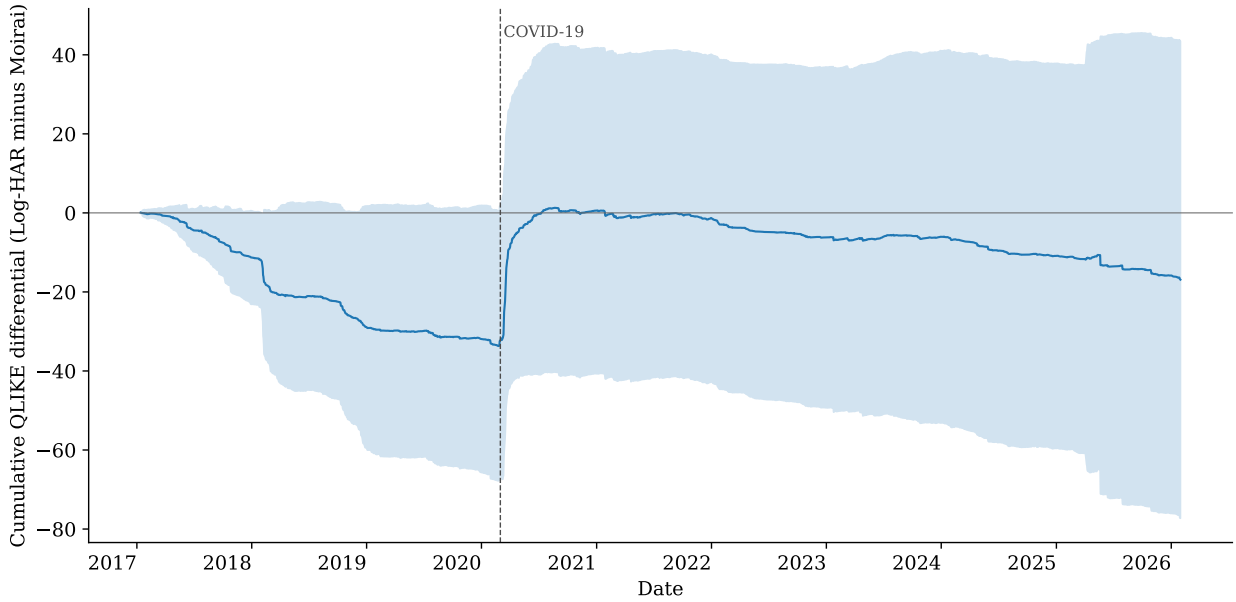

Figure 5: Cumulative QLIKE loss differential (Log-HAR minus Moirai 2.0) at $h = 1$, averaged across 40 equities. Positive values indicate that Moirai 2.0 outperforms Log-HAR. The shaded area is ± 1 cross-sectional standard deviation. The vertical dashed line marks March 2020 (COVID-19 onset).

Forecast efficiency: MZ regressions. Table 11 reports average MZ regression results across 50 assets. The MZ regression, $RV_t = \alpha + \beta \hat{f}_t + \varepsilon_t$, tests whether forecasts are efficient: under the null, $\alpha = 0$ and $\beta = 1$.

At $h = 1$, Moirai 2.0 and ARFIMA produce the most well-calibrated forecasts ($\hat{\beta} = 1.05$ for both), closest to the theoretical ideal of unity. Sundial ($\hat{\beta} = 1.14$) and TimesFM 2.5 ($\hat{\beta} = 1.18$) also remain close to unity. The HAR family overshoots modestly ($\hat{\beta} = 1.29$ for HAR, 1.25 for HAR-J), while Log-HAR overshoots substantially ($\hat{\beta} = 1.77$), reflecting the bias-correction step that converts log-space forecasts back to levels. Among TSFMs, TimesFM 2.5 achieves the highest MZ R^2 (0.432), followed by Chronos-Bolt-B (0.423) and Moirai-MoE-S (0.415). Two new models show systematic undershooting: Moirai-MoE-S ($\hat{\beta} = 0.79$) and Toto ($\hat{\beta} = 0.44$), indicating compressed forecasts. The joint efficiency null ($\alpha = 0$, $\beta = 1$) is rejected for 90–100% of assets for all TSFMs, compared to 54–58% for

HAR and HAR-J. This seemingly paradoxical result (TSFMs are more accurate on MSE and R^2 but more frequently reject efficiency) arises because the MZ test has greater power when R^2 is high.

At $h = 22$ (Table 18), all econometric models exhibit substantial undershoot ($\hat{\beta} = 0.30$–0.58), with MZ R^2 falling to 0.07–0.14. TSFMs retain substantially better calibration: Moirai 2.0 achieves $\hat{\beta} = 0.99$, Sundial $\hat{\beta} = 0.89$, and TimesFM 2.5 the highest R^2 (0.235). Moirai-MoE-S undershoots more at this horizon ($\hat{\beta} = 0.48$), while Toto's severe undershoot ($\hat{\beta} = 0.26$) confirms its persistent scale misalignment. This confirms that the best TSFMs' advantage at longer horizons is not only in loss-function terms but also in the structural quality of their forecasts.

Table 11: Mincer–Zarnowitz forecast efficiency regression ($h = 1$). Cross-sectional averages across 50 assets (40 equities, 5 FX, 5 futures). The regression is $RV_t = \alpha + \beta \hat{f}_t + \varepsilon_t$; under forecast efficiency, $\alpha = 0$ and $\beta = 1$. The F-statistic tests the joint null with HAC (Newey–West) standard errors. % Reject reports the fraction of assets for which the joint null is rejected at the 5% level.

Model	$\hat{\alpha}$	$\hat{\beta}$	R^2	F-stat	p(joint)	% Reject
Econometric benchmarks						
HAR	−0.0000	1.288	0.365	20.69	0.154	58%
HAR-J	−0.0000	1.245	0.374	11.02	0.146	54%
HAR-RS	0.0000	1.123	0.309	24.11	0.099	72%
HARQ	0.0001	0.893	0.142	13.52	0.028	82%
Log-HAR	−0.0001	1.770	0.308	8.25	0.022	90%
ARFIMA	0.0000	1.054	0.171	18.93	0.059	84%
Foundation models						
Chronos-Bolt-S	0.0000	1.200	0.402	34.50	0.004	98%
Chronos-Bolt-B	0.0000	1.253	0.423	42.74	0.003	98%
Moirai-2.0-S	0.0000	1.050	0.378	22.17	0.002	98%
Lag-Llama	−0.0001	1.499	0.147	7.21	0.041	76%
TimesFM-2.5	0.0000	1.183	0.432	33.89	0.004	98%
Toto	0.0001	0.442	0.257	44.23	0.000	100%
Sundial	0.0000	1.138	0.412	14.94	0.014	90%
Moirai-MoE-S	0.0001	0.789	0.415	28.21	0.001	100%

Time-varying performance: GR fluctuation test. Figure 6 plots the rolling DM statistic (QLIKE loss) of each model against HAR at $h = 1$, averaged across 50 assets. A negative rolling DM indicates that the comparison model produces lower QLIKE than HAR during that window; values beyond the ± 2.80 critical-value bands indicate rejection of the equal predictive ability null at the 5% level.

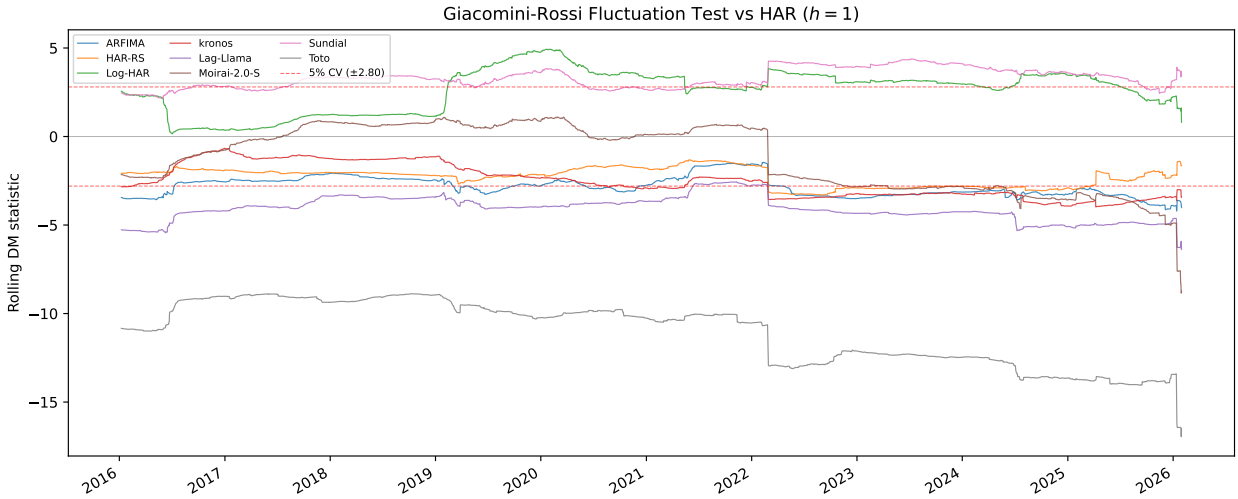

Figure 6: GR fluctuation test: rolling DM statistic (QLIKE loss) of each model vs. HAR at $h = 1$, averaged across 50 assets. Negative values indicate the comparison model outperforms HAR. Dashed red lines mark the 5% critical values (± 2.80). The window size is 30% of the sample.

Three patterns emerge. First, the relative performance of all models against HAR is time-varying: the null of constant relative predictive ability is rejected for 100% of assets for Log-HAR, ARFIMA, Lag-Llama, Sundial, and Toto, and for 62–94% for the remaining TSFMs. This means that unconditional DM tests and MCS results, while informative about average performance, mask substantial temporal variation. Second, Log-HAR's advantage over HAR is concentrated in the 2019–2021 period, where it crosses above the upper critical value, indicating that it significantly *underperforms* HAR during the COVID-induced volatility spike. This is consistent with the log-space bias correction amplifying forecast errors during regime changes. After 2022, Log-HAR's advantage narrows and approaches zero. Third,

TSFMs (which enter the sample around 2022 when sufficient context becomes available) immediately settle below the lower critical-value band, indicating persistent and significant outperformance of HAR. Sundial shows the most stable advantage, with 100% rejection at $h = 1$.

Figures 7 and 8 present the corresponding plots at $h = 5$ and $h = 22$. At $h = 22$, the fluctuation test rejects less frequently (24–86% across models), consistent with the finding that forecast differences diminish at longer horizons as all models converge toward the unconditional mean (Patton, 2011).

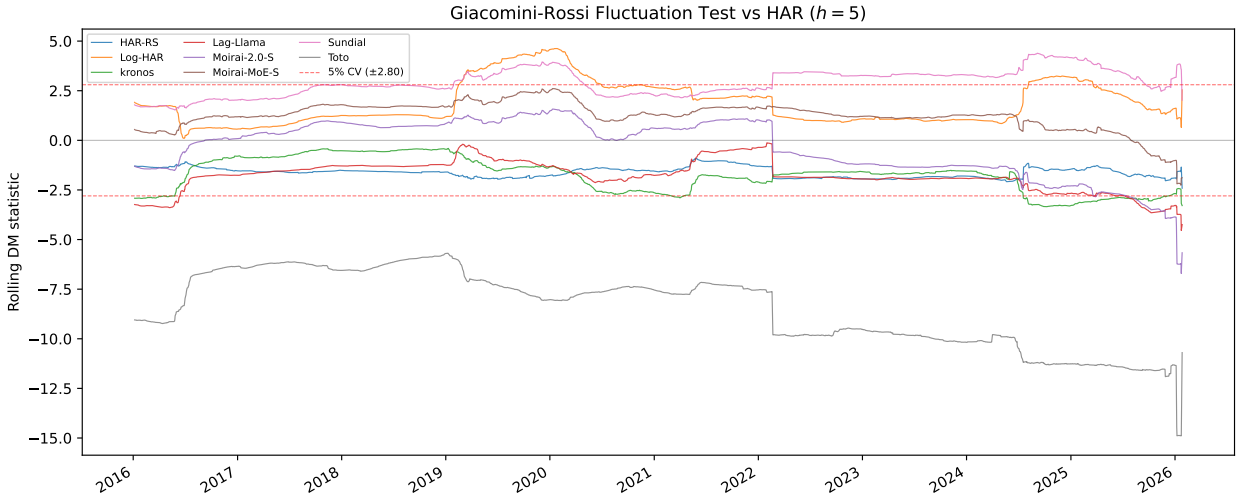

Figure 7: GR fluctuation test at $h = 5$. Each panel plots the rescaled cumulative loss differential between the column model and HAR, with ±5% critical values (dashed).

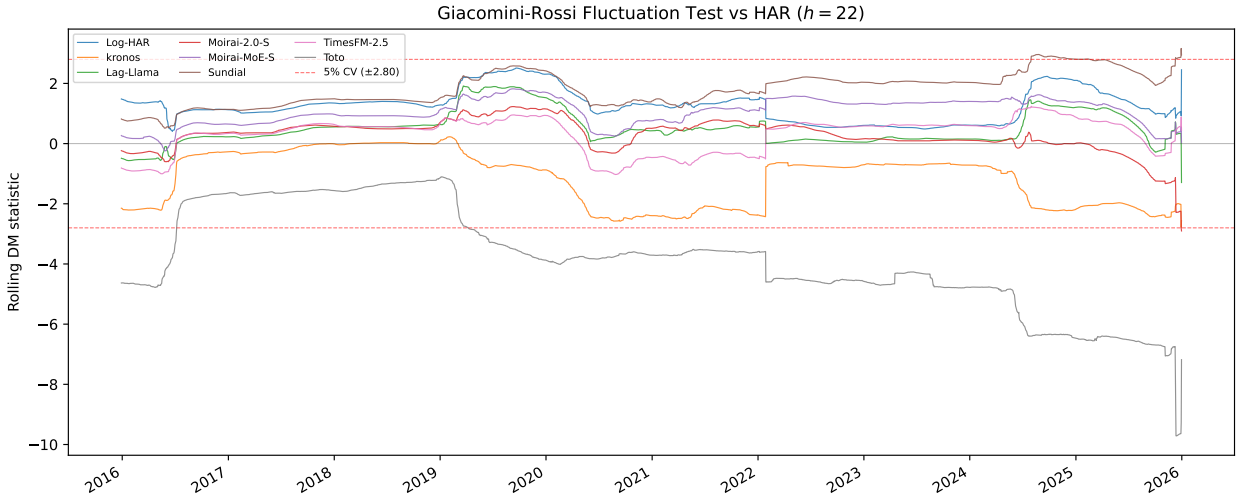

Figure 8: GR fluctuation test at $h = 22$. Each panel plots the rescaled cumulative loss differential between the column model and HAR, with ±5% critical values (dashed).

Interpretation. Taken together, the four tests support the same conclusion: the best TSFMs produce statistically superior and better-calibrated forecasts than HAR-family models, but the TSFM class is not uniformly strong. The MCS identifies Sundial as statistically dominant at all horizons, with Moirai-MoE-S as a consistent runner-up. The DM tests confirm that Sundial's win rate (87.8% at $h = 1$) far exceeds all other models. The MZ regressions show that the best TSFMs maintain calibration at long horizons ($\hat{\beta}$ closer to 1), while econometric models suffer severe undershoot at $h = 22$. The GR test reveals that relative performance is not constant: Log-HAR's advantage erodes over time, and TSFMs consistently outperform HAR from their entry into the sample onward. The wide heterogeneity across TSFM architectures—from Sundial's dominance to Toto's severe miscalibration—suggests that pretraining data composition and output mechanism (flow matching vs. quantile regression) matter as much as the zero-shot paradigm itself.

7 Robustness Checks

We assess the stability of our main findings along several dimensions.

Sub-sample stability: pre- and post-COVID. Table 12 splits the equity sample at March 1, 2020, and reports forecast accuracy separately for the pre-COVID (2015–2020) and post-COVID (2020–2026) periods. The sub-sample analysis reveals that TSFM performance is regime-robust, while levels-based econometric models are not.

In the pre-COVID period, Sundial achieves the lowest QLIKE at $h = 1$ (0.198) and $h = 5$ (0.179), while Log-HAR leads at $h = 22$ (0.320). The pre-COVID period features lower average volatility and fewer extreme events, yet Sundial already outperforms all econometric models. Toto remains severely miscalibrated in both periods (QLIKE 34.1 pre-COVID, 34.8 post-COVID at $h = 1$).

In the post-COVID period, TSFMs dominate across all metrics and horizons. Sundial achieves the lowest QLIKE at $h = 1$ (0.193 vs. 0.224 for Log-HAR), $h = 5$ (0.152 vs. 0.249), and $h = 22$ (0.135 vs. 0.329). TimesFM 2.5 achieves the highest R^2_{OOS} at $h = 1$ (0.445), while Sundial leads at $h = 5$ (0.386) and $h = 22$ (0.192). The levels-based HAR variants suffer extreme QLIKE inflation in the high-volatility regime: HARQ's QLIKE rises from 2.35 pre-COVID to 22.98 post-COVID at $h = 1$, because the near-zero forecast problem is exacerbated when realized values are large. Log-HAR is more stable across regimes than its levels-based counterparts but still falls behind the TSFMs on all metrics in the post-COVID sample. The TSFM heterogeneity pattern persists across sub-samples: Lag-Llama underperforms in both periods at $h = 1$ (QLIKE 0.433 pre-COVID, 0.385 post-COVID) but improves at longer horizons. These results suggest that TSFMs' advantage is present in both calm and volatile market conditions, though the gap widens in the post-COVID regime where their flexible nonparametric forecasts adapt more effectively than the HAR's linear structure.

Estimation window: 252 vs. 512 days. A potential concern is that TSFMs receive 512 days of context while econometric models use 252-day estimation windows, creating an information asymmetry that may favor TSFMs. To address this, we re-estimate all

Table 12: Sub-sample stability: pre-COVID (2015–2020) and post-COVID (2020–2026) forecast accuracy for 40 U.S. equities (VOLARE). Bold indicates best in each column within each panel. † denotes QLIKE values inflated by near-zero forecasts.

Model	MSE ($\times 10^{-6}$)	MAE ($\times 10^{-4}$)	QLIKE	R^2_{OOS}
Panel: $h = 1$, Pre-COVID				
HAR	0.088	0.973	0.493	0.229
HAR-J	0.089	0.964	0.393	0.209
HAR-RS	0.100	1.025	1.261†	0.168
HARQ	0.090	0.950	2.351†	0.177
Log-HAR	0.086	0.854	0.206	0.246
ARFIMA	0.118	1.122	0.334	-0.008
Chronos-Bolt-S	0.086	0.834	0.235	0.247
Chronos-Bolt-B	0.084	0.823	0.234	0.275
Moirai-2.0-S	0.084	0.857	0.251	0.235
Lag-Llama	0.106	1.188	0.433	0.007
TimesFM-2.5	0.083	**0.818**	0.229	0.274
Toto	0.131	1.960	34.082†	-1.405
Sundial	**0.081**	0.849	**0.198**	**0.301**
Moirai-MoE-S	0.086	0.862	0.217	0.239
Panel: $h = 1$, Post-COVID				
HAR	0.327	1.679	1.808†	0.335
HAR-J	0.322	1.661	1.473†	0.349
HAR-RS	0.355	1.784	5.012†	0.264
HARQ	0.405	1.775	22.979†	0.096
Log-HAR	0.346	1.541	0.224	0.234
ARFIMA	0.399	1.897	0.321	0.118
Chronos-Bolt-S	0.285	1.363	0.226	0.410
Chronos-Bolt-B	0.281	1.353	0.228	0.422
Moirai-2.0-S	0.277	1.360	0.213	0.430
Lag-Llama	0.388	1.899	0.385	0.120
TimesFM-2.5	**0.274**	**1.339**	0.218	**0.445**
Toto	0.390	2.647	34.844†	0.042
Sundial	0.278	1.402	**0.193**	0.432
Moirai-MoE-S	0.281	1.421	0.207	0.423
Panel: $h = 5$, Pre-COVID				
HAR	0.048	0.960	0.420	0.037
HAR-J	0.050	0.953	0.426	0.024
HAR-RS	0.060	1.016	2.050†	-0.090
HARQ	0.048	0.943	1.813†	0.043
Log-HAR	**0.042**	0.829	0.182	0.197
ARFIMA	0.060	0.997	0.268	-0.064
Chronos-Bolt-S	0.047	0.791	0.241	0.168
Chronos-Bolt-B	0.046	0.782	0.240	0.187
Moirai-2.0-S	0.045	0.789	0.217	0.202
Lag-Llama	0.056	1.024	0.274	0.026
TimesFM-2.5	0.046	**0.776**	0.238	0.187
Toto	0.063	1.320	19.974†	-0.625
Sundial	0.043	0.783	**0.179**	**0.247**
Moirai-MoE-S	0.044	0.777	0.190	0.234
Panel: $h = 5$, Post-COVID				
HAR	0.219	1.633	3.872†	0.201
HAR-J	0.213	1.611	3.520†	0.227
HAR-RS	0.247	1.763	9.314†	0.087
HARQ	0.263	1.683	23.111†	0.002
Log-HAR	0.223	1.487	0.249	0.125
ARFIMA	0.235	1.674	0.245	0.120
Chronos-Bolt-S	0.170	1.256	0.200	0.348
Chronos-Bolt-B	0.172	1.261	0.204	0.337
Moirai-2.0-S	0.162	**1.216**	0.174	0.378
Lag-Llama	0.246	1.638	0.321	0.020
TimesFM-2.5	0.162	1.217	0.192	0.381
Toto	0.224	1.848	14.155†	0.020
Sundial	**0.161**	1.251	**0.152**	**0.386**
Moirai-MoE-S	0.163	1.245	0.156	0.364

six econometric benchmarks with a 512-day rolling window, matching the TSFM context length. Table 13 reports the results at $h = 1$ for the 40-equity sample. Five of six models improve under the longer window: HAR R^2_{OOS} rises from 0.340 to 0.435, and levels-HAR QLIKE inflation decreases substantially (e.g., HAR-RS QLIKE drops from 3.48 to 0.57). The exception is ARFIMA, whose long-memory estimation degrades with a 512-day window (R^2_{OOS} falls from 0.132 to -0.332). Cross-model R^2_{OOS} rankings are highly stable (Spearman $\rho = 0.94$, $p < 0.01$). Importantly, even after granting econometric models the same 512-day information set, the best TSFM zero-shot QLIKE values (Sundial at 0.195, Moirai-MoE-S at 0.211; Table 5) remain competitive with or below the best 512-day econometric QLIKE (Log-HAR at 0.20), confirming that the TSFM advantage is not an artifact of the longer context window. The same pattern holds at longer horizons: Tables 19 and 20 in Appendix A confirm that the 512-day window improves all HAR variants at $h = 5$ and $h = 22$, while ARFIMA deteriorates further.

Table 13: Estimation window sensitivity: 252 vs. 512 days. Estimation window sensitivity: 252-day vs. 512-day rolling window ($h = 1$, 40 equities). Cross-asset mean metrics. Bold indicates the better window for each model.

	MSE ($\times 10^8$)		QLIKE		R^2_{OOS}	
Model	252	512	252	512	252	512
HAR	23.03	**21.38**	1.2715	**0.2107**	0.340	**0.435**
HAR-J	22.78	**20.89**	1.0356	**0.3590**	0.353	**0.442**
HAR-RS	25.18	**21.92**	3.4828	**0.5729**	0.272	**0.417**
HARQ	27.71	**24.64**	14.5773	**4.8653**	0.119	**0.270**
Log-HAR	23.98	**22.46**	0.2166	**0.1954**	0.249	**0.381**
ARFIMA	**28.39**	39.98	**0.3264**	0.4786	**0.132**	-0.332

Context window length. Table 14 examines how TSFM forecast accuracy varies with context window length (128, 256, and 512 days). Longer context generally improves Chronos-Bolt: QLIKE decreases from 0.231 (ctx=128) to 0.222 (ctx=512) at $h = 1$, with similar patterns at $h = 5$. Moirai 2.0 shows a more nuanced pattern: while longer context improves QLIKE at $h = 1$ (0.264 to 0.243), it *worsens* performance at $h = 22$ (0.239 at ctx=128 vs.

0.251 at ctx=512), suggesting that Moirai's architecture captures long-range dependencies efficiently even from shorter windows and that additional context introduces noise at the monthly horizon. Lag-Llama benefits substantially from longer context at $h = 5$ and $h = 22$ (QLIKE drops from 0.515 to 0.297 at $h = 5$), but at $h = 1$ the relationship is non-monotonic, with the best QLIKE at ctx=256 (0.380). These results confirm that the 512-day default is appropriate for Chronos-Bolt but not universally optimal across architectures and horizons.

Table 14: Context-length sensitivity of TSFM forecasts. QLIKE and R^2_{OOS} averaged across all assets, by horizon and context window length. Bold indicates the best context length for each model–horizon pair.

		QLIKE			R^2_{OOS}		
Model	h	ctx=128	ctx=256	ctx=512	ctx=128	ctx=256	ctx=512
Chronos-Bolt-S	1	0.231	0.232	**0.222**	0.337	0.330	**0.378**
	5	0.216	0.208	**0.203**	0.259	0.278	**0.345**
	22	**0.299**	0.305	0.303	-0.183	0.089	**0.109**
Chronos-Bolt-B	1	0.401	0.245	**0.224**	0.369	0.377	**0.389**
	5	0.346	**0.206**	0.206	0.329	0.317	**0.337**
	22	0.394	**0.302**	0.307	0.087	0.098	**0.102**
Moirai-2.0-S	1	0.264	0.249	**0.243**	**0.364**	0.347	0.362
	5	0.201	0.199	**0.197**	**0.358**	0.319	0.331
	22	**0.239**	0.243	0.251	**0.149**	0.112	0.098
Lag-Llama	1	0.433	**0.380**	0.464	**0.191**	0.168	0.117
	5	0.515	0.312	**0.297**	0.027	**0.061**	0.054
	22	0.587	0.332	**0.276**	-0.076	0.003	**0.027**

QLIKE floor sensitivity. Our QLIKE implementation floors forecasts at 10^{-10} to prevent division by zero. Because levels-based HAR variants occasionally produce near-zero forecasts, this floor choice could affect model rankings. Table 15 reports cross-asset mean QLIKE at $h = 1$ under four floor values (10^{-4}, 10^{-6}, 10^{-8}, 10^{-10}). QLIKE values are identical for floors $\leq 10^{-6}$ (Spearman rank correlation $\rho = 1.0$ vs. 10^{-10}), confirming that the floor never binds at these levels. Only the aggressive 10^{-4} floor alters rankings ($\rho = 0.82$) by clipping levels-HAR forecasts that drive QLIKE inflation (e.g., HARQ drops from 16.04 to 0.38). Our default 10^{-10} floor is conservative and does not distort model comparisons.

Table 15: QLIKE floor sensitivity: cross-asset mean QLIKE at $h = 1$ under four floor values. Rankings are stable across floors (Spearman $\rho > 0.99$ for floors $\leq 10^{-6}$).

Model	10^{-10}	10^{-8}	10^{-6}	10^{-4}
HAR	1.38	1.38	1.38	0.2565
HARQ	16.04	16.04	16.04	0.3827
ARFIMA	0.3218	0.3218	0.3218	0.3195
Log-HAR	0.2189	0.2189	0.2189	0.2358
HAR-J	1.11	1.11	1.11	0.2461
HAR-RS	3.63	3.63	3.63	0.2838
Toto	34.59	34.59	34.59	0.4485
Chronos-Bolt-S	0.2292	0.2292	0.2292	0.2353
Lag-Llama	0.4018	0.4018	0.4018	0.3639
Sundial	0.1951	0.1951	0.1951	0.2121
Chronos-Bolt-B	0.2302	0.2302	0.2302	0.2351
Moirai-2.0-S	0.2263	0.2263	0.2263	0.2274
Moirai-MoE-S	0.2106	0.2106	0.2106	0.2203
TimesFM-2.5	0.2221	0.2221	0.2221	0.2287

MZ bias-corrected evaluation. A model's low loss could reflect good calibration (i.e., alignment between the scale of forecasts and realizations) rather than genuine predictive information about future volatility dynamics. The MZ regression provides a natural correction: by regressing realized values on forecasts, one obtains intercept and slope parameters that remove any linear bias. If a model's accuracy survives this correction, its forecasts contain information beyond what a simple affine rescaling could produce. To test this, we apply a recursive MZ bias correction to all TSFM forecasts at $h = 1$: at each time t, we estimate $\hat{\alpha}_t$ and $\hat{\beta}_t$ from all observations up to $t - 1$, then form the corrected forecast $\hat{f}_t^{\mathrm{MZ}} = \hat{\alpha}_t + \hat{\beta}_t f_t$ using an expanding estimation window (minimum 252 days). Table 16 compares original and MZ-corrected TSFM forecasts against econometric benchmarks, evaluated on the common post-warm-up sample. MZ correction improves QLIKE for most well-calibrated models (TimesFM 2.5: 0.2221 $\rightarrow$ 0.1887; Chronos-Bolt-Base: 0.2307 $\rightarrow$ 0.1918), while Sundial's already-optimal QLIKE is essentially unchanged (0.1950 $\rightarrow$ 0.1945). MZ correction dramatically improves Toto (35.69 $\rightarrow$ 3.94), indicating that much of Toto's poor performance stems from severe scale misalignment rather than absence of signal. Lag-Llama worsens

under correction, where the linear adjustment pushes some forecasts toward zero. Moirai-MoE-S improves modestly (0.2098 → 0.2038), while Moirai 2.0 is nearly unchanged (0.2245 → 0.2238), indicating that both Moirai variants are already well calibrated at this horizon. MSE and R^2_{OOS} are largely unchanged for the well-calibrated models, confirming that TSFM accuracy reflects genuine predictive content rather than calibration artifacts.

Table 16: Mincer–Zarnowitz bias-corrected TSFM evaluation (cross-asset mean, $h = 1$). Each TSFM forecast $\hat{f}_t$ is replaced by $\hat{\alpha}+\hat{\beta}\hat{f}_t$ from a rolling Mincer–Zarnowitz regression, then re-evaluated on MSE, QLIKE, and R^2_{OOS}. Econometric benchmarks are shown unchanged for comparison.

Model	Variant	MSE ($\times 10^8$)	QLIKE	R^2_{OOS}
HAR	Original	27.07	1.52	0.333
HAR-J	Original	26.76	1.22	0.347
HAR-RS	Original	29.44	4.04	0.266
HARQ	Original	32.98	17.97	0.105
Log-HAR	Original	28.41	0.2220	0.237
ARFIMA	Original	33.34	0.3279	0.120
Chronos-Bolt-S	Original	24.04	0.2298	0.403
Chronos-Bolt-S	MZ-Corrected	27.51	0.1969	0.322
Chronos-Bolt-B	Original	23.67	0.2307	0.415
Chronos-Bolt-B	MZ-Corrected	25.40	0.1918	0.383
Moirai-2.0-S	Original	23.37	0.2245	0.421
Moirai-2.0-S	MZ-Corrected	25.56	0.2238	0.331
Lag-Llama	Original	32.36	0.4162	0.117
Lag-Llama	MZ-Corrected	32.46	7.05	0.116
TimesFM-2.5	Original	23.13	0.2221	0.437
TimesFM-2.5	MZ-Corrected	24.94	0.1887	0.395
Toto	Original	32.96	35.69	0.003
Toto	MZ-Corrected	28.62	3.94	0.250
Sundial	Original	23.35	0.1950	0.425
Sundial	MZ-Corrected	24.75	0.1945	0.388
Moirai-MoE-S	Original	23.76	0.2098	0.414
Moirai-MoE-S	MZ-Corrected	26.44	0.2038	0.369

8 Conclusion

This paper evaluates eight zero-shot time series foundation models for realized volatility forecasting across 50 assets spanning U.S. equities, foreign exchange, and futures.

Three main findings emerge. First, the best zero-shot TSFMs outperform the best econometric benchmarks for univariate realized variance forecasting, contrasting with prior negative findings on daily returns and on TimesFM 1.0. Sundial, a flow-matching generative model, achieves the lowest QLIKE across all three asset classes and all horizons, entering the Model Confidence Set for 97.5–100% of equities. However, performance is highly heterogeneous across TSFM architectures: Toto produces severely miscalibrated forecasts (QLIKE exceeding 6 at every horizon), though MZ bias correction reveals underlying signal, while Sundial, Moirai-MoE-S, and TimesFM 2.5 match or exceed the best econometric benchmarks. Second, the advantage of the top-performing TSFMs grows with forecast horizon: at $h = 1$, Sundial's QLIKE (0.195) leads Log-HAR (0.219) by 11%, but at $h = 22$ the gap widens to 37% (0.218 vs. 0.345). This pattern holds for Moirai-MoE-S as well, though not for all TSFMs. Third, rankings within the TSFM class are more variable than across classes: Sundial dominates equities, FX, and futures alike, while the relative ordering of other TSFMs (Moirai-MoE-S, Moirai 2.0, TimesFM 2.5, Chronos-Bolt) shifts across asset classes.

Several limitations should be noted. We evaluate only zero-shot TSFM performance; parameter-efficient fine-tuning may yield further gains, though the adaptation mechanisms differ across architectures (Chronos-2 supports LoRA-based adaptation through its encoder-decoder structure, while Lag-Llama uses epoch-based training through its GluonTS estimator interface). Our evaluation is restricted to point forecasts; density forecast evaluation remains an open question. VOLARE provides alternative realized measures (subsampled realized variance, realized kernels, range-based estimators) whose effect on model rankings we leave for future work. Extension to realized covariance forecasting and portfolio construction is a natural next step. While no TSFM in our study was trained on realized volatility (Section 4.3), indirect exposure to related financial series such as exchange rates

and macroeconomic indicators cannot be fully ruled out as a contributing factor; future work with held-out asset classes or temporal holdouts could further address this concern.

For practitioners, three concrete recommendations emerge. First, Sundial is the recommended default for zero-shot realized volatility forecasting: it dominates on QLIKE across all asset classes and horizons, enters the MCS for 97.5–100% of equities, and its already near-optimal calibration means that post-hoc MZ bias correction yields negligible further improvement. Second, for monthly-horizon forecasting ($h = 22$), TSFMs should replace HAR as the default benchmark, since econometric models barely outperform the historical mean ($R^2_{\text{OOS}} < 0$ for Log-HAR) while Sundial maintains $R^2_{\text{OOS}} = 0.148$. Third, MZ bias correction can rescue underperforming TSFMs—Toto's QLIKE drops from 34.59 to 3.94 after rescaling, revealing underlying signal beneath its severe miscalibration—suggesting that a simple post-processing step may extend the usable TSFM set.

More broadly, the finding that models pretrained on weather, energy, and retail data can forecast financial volatility without any domain-specific training raises questions about the nature of temporal patterns that transfer across domains. Realized volatility shares key statistical properties—positivity, mean reversion, long memory, right-skewed distributions—with many physical and economic processes, and TSFMs appear to exploit these shared structures. At the same time, the wide performance dispersion across architectures cautions against treating TSFMs as a uniform class. Model selection within the TSFM family is as important as the choice between TSFMs and econometric models.

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A Supplementary Tables

This appendix collects additional results referenced in the main text. Tables A, A, and A provide per-asset QLIKE breakdowns at $h = 1$, $h = 5$, and $h = 22$ across all 50 assets. Tables 17 and 18 present Mincer–Zarnowitz regressions at $h = 5$ and $h = 22$. Tables 19 and 20 extend the estimation window sensitivity analysis to $h = 5$ and $h = 22$.

Asset	HAR	Log-HAR	ARFIMA	Chr-Bolt-S	Chr-Bolt-B	Moirai-S	Lag-Llama	TFM-2.5	Toto	Sundial	MoE-S
Stocks											
AAPL	0.1997	**0.1932**	0.4700	0.2278	0.2328	0.2067	0.4122	0.2225	38.0542	0.1937	0.2209
ADBE	0.1570	**0.1507**	0.4327	0.1760	0.1794	0.1621	0.3468	0.1749	36.5618	0.1516	0.1601
AMD	0.1532	**0.1500**	0.2883	0.1697	0.1725	0.1633	0.2888	0.1660	8.2771	0.1509	0.1717
AMGN	0.2145	0.2106	0.3082	0.2409	0.2351	0.2193	0.3873	0.2311	37.4056	**0.2010**	0.2100
AMZN	13.8320	0.4528	0.6815	0.5050	0.5127	0.4790	0.8656	0.4895	34.1204	**0.4157**	0.4676
AXP	0.2137	0.2241	0.2973	0.2111	0.2106	0.1959	0.4140	0.2020	37.9825	**0.1762**	0.1894
BA	9.3426	0.2843	0.3670	0.2439	0.2447	0.2248	0.4943	0.2406	32.5503	**0.2041**	0.2326
CAT	0.1769	0.1793	0.2530	0.1895	0.1907	0.1693	0.3422	0.1850	38.2613	**0.1615**	0.1735
CRM	0.2299	0.2170	0.3759	0.2516	0.2520	0.2341	0.3983	0.2413	34.5271	**0.2072**	0.2232
CSCO	0.1911	0.1852	0.3172	0.2102	0.2098	0.1913	0.3785	0.2022	35.9292	**0.1798**	0.1936
CVX	0.1870	0.1962	0.2593	0.1583	0.1566	0.1658	0.3065	0.1540	35.5364	**0.1414**	0.1468
DIS	2.5596	0.2216	0.3013	0.2302	0.2345	0.2058	0.4009	0.2242	38.6116	**0.1937**	0.2084
GE	0.2007	0.1949	0.2884	0.2173	0.2185	0.2025	0.3640	0.2104	29.8779	**0.1850**	0.2008
GOOGL	0.3662	0.3088	0.4195	0.3416	0.3519	0.3140	0.5129	0.3408	42.1751	**0.2902**	0.3223
GS	0.1670	0.1643	0.2697	0.1739	0.1715	0.1529	0.3189	0.1661	40.4164	**0.1505**	0.1612
HD	0.1971	0.1975	0.2923	0.1957	0.1939	0.1836	0.4202	0.1847	41.1524	**0.1620**	0.1837
HON	0.2256	0.2227	0.3146	0.2078	0.2058	0.2115	0.4032	0.1959	37.3624	**0.1738**	0.1826
IBM	0.6798	0.1981	0.3104	0.2133	0.2157	0.2109	0.3832	0.2044	37.7081	**0.1823**	0.1879
JNJ	0.2310	0.2194	0.2925	0.2152	0.2128	0.2529	0.3978	0.2095	33.4378	**0.1892**	0.2006
JPM	0.1874	0.1848	0.3035	0.1950	0.1943	0.1768	0.3618	0.1883	35.8822	**0.1648**	0.1755
KO	0.1827	0.1767	0.2511	0.1675	0.1655	0.2524	0.3446	0.1570	28.8196	**0.1435**	0.1493
MCD	3.9225	0.2386	0.2976	0.2252	0.2259	0.2622	0.3831	0.2192	31.4435	**0.1945**	0.2056
META	0.2198	0.1855	0.3188	0.1887	0.1951	0.1790	0.3721	0.1876	35.4874	**0.1649**	0.1766
MMM	0.2185	0.2121	0.2965	0.2271	0.2280	0.2509	0.3729	0.2232	43.3445	**0.1916**	0.2073
MRK	0.2064	0.2000	0.2678	0.2212	0.2184	0.1922	0.3507	0.2073	36.7627	**0.1809**	0.1962
MSFT	0.2980	0.2718	0.4233	0.3311	0.3256	0.3193	0.4759	0.3122	41.9633	**0.2679**	0.2891
NFLX	10.5396	0.3476	0.4306	0.3809	0.3801	0.3696	0.5480	0.3673	21.9215	**0.3224**	0.3658
NKE	0.2170	0.2098	0.2835	0.2105	0.2125	0.1818	0.3396	0.2019	41.4459	**0.1805**	0.1897
NVDA	0.1909	0.1827	0.3225	0.2017	0.2062	0.1907	0.3857	0.1995	19.9380	**0.1753**	0.1927
ORCL	1.6483	0.2585	0.3809	0.2854	0.2961	0.2608	0.5327	0.2806	36.6540	**0.2368**	0.2701
PG	0.2001	0.1900	0.2404	0.1812	0.1806	0.2332	0.3771	0.1740	28.2630	**0.1572**	0.1630
PM	0.2589	0.2424	0.2899	0.2549	0.2589	0.2433	0.4061	0.2558	41.6249	**0.2280**	0.2349
SHW	1.5060	0.2060	0.2484	0.1991	0.2005	0.1784	0.3264	0.1922	42.8356	**0.1719**	0.1797
TRV	0.1689	0.1655	0.2349	0.1726	0.1691	0.1858	0.3279	0.1650	33.9876	**0.1492**	0.1556
TSLA	2.2254	0.1722	0.2929	0.2029	0.2036	0.1869	0.3343	0.1974	13.8548	**0.1697**	0.1967
UNH	0.2226	0.2147	0.3134	0.2519	0.2527	0.2228	0.4056	0.2401	38.4409	**0.2080**	0.2269
V	0.8008	0.2616	0.3140	0.2686	0.2732	0.3309	0.4384	0.2644	33.9561	**0.2373**	0.2393
VZ	1.1632	0.1859	0.2658	0.1998	0.2035	0.2040	0.3847	0.1945	34.9943	**0.1786**	0.1888
WMT	0.9400	0.2522	0.3115	0.2517	0.2498	0.2743	0.4637	0.2481	35.1741	**0.2187**	0.2280
XOM	0.2045	0.2264	0.2449	0.1709	0.1678	0.2122	0.3034	0.1615	36.7170	**0.1511**	0.1559
Average	1.3761	0.2189	0.3218	0.2292	0.2302	0.2263	0.4018	0.2221	34.5865	**0.1951**	0.2106
FX											
AUDUSD	0.1617	0.1524	0.2132	0.1504	0.1536	0.3086	0.2664	0.1482	16.6629	**0.1366**	0.1487
EURUSD	0.1571	0.1491	0.2035	0.1576	0.1594	0.3139	0.2768	0.1560	11.1293	**0.1393**	0.1988
GBPUSD	0.1504	0.1376	0.2120	0.1492	0.1487	0.3198	0.2806	0.1437	12.8796	**0.1327**	0.1667
USDCAD	0.1380	0.1308	0.1749	0.1326	0.1331	0.2587	0.2376	0.1310	9.5052	**0.1229**	0.1824
USDJPY	0.2753	0.2435	0.4040	0.2823	0.2856	0.4456	0.4621	0.2754	13.0623	**0.2350**	0.2968
Average	0.1765	0.1627	0.2415	0.1744	0.1761	0.3293	0.3047	0.1709	12.6479	**0.1533**	0.1987
Futures											
C	0.3153	0.2903	0.3282	0.3256	0.3269	0.2969	0.5447	0.3200	37.1299	**0.2765**	0.3023
CL	0.1998	0.1358	0.3310	0.1591	0.1521	0.1741	3.0510	0.1363	24.1237	0.1483	**0.1358**
ES	0.2286	0.1976	0.5399	0.2244	0.2224	0.5307	1.2473	0.2108	24.6635	**0.1927**	0.2222
GC	0.1648	**0.1601**	0.2879	0.1763	0.2223	0.2669	0.3741	0.1734	28.8041	0.1612	0.2117
NG	0.2176	0.1787	0.2421	0.1790	0.1802	0.1773	0.3715	0.1785	9.7896	**0.1654**	0.1824
Average	0.2252	0.1925	0.3458	0.2129	0.2208	0.2892	1.1177	0.2038	24.9021	**0.1888**	0.2109
Overall Average	1.1411	0.2106	0.3162	0.2221	0.2239	0.2429	0.4637	0.2151	31.4242	**0.1903**	0.2094

Per-asset QLIKE loss at $h = 1$. QLIKE is computed as $\text{mean}(RV_t/\hat{RV}_t - \ln(RV_t/\hat{RV}_t) - 1)$. Bold indicates the lowest (best) QLIKE in each row. Chr-Bolt-S = Chronos-Bolt-Small, Chr-Bolt-B = Chronos-Bolt-Base, Moirai-S = Moirai-2.0-Small. HAR-J, HAR-RS, and HARQ are omitted as their QLIKE values are dominated by near-zero forecast artifacts (see Section 5.1).

Asset	HAR	Log-HAR	ARFIMA	Chr-Bolt-S	Chr-Bolt-B	Moirai-S	Lag-Llama	TFM-2.5	Toto	Sundial	MoE-S
Stocks											
AAPL	0.1783	**0.1668**	0.3870	0.2314	0.2385	0.1927	0.2922	0.2280	20.1222	0.1728	0.1757
ADBE	0.1264	**0.1173**	0.3805	0.1643	0.1699	0.1406	0.2420	0.1608	13.5529	0.1235	0.1308
AMD	0.1108	0.1086	0.2138	0.1389	0.1430	0.1336	0.1725	0.1353	0.4828	**0.1073**	0.1301
AMGN	0.2248	0.2053	0.2356	0.2363	0.2345	0.2044	0.3123	0.2290	18.5348	**0.1801**	0.1818
AMZN	19.2609	0.4705	0.5729	0.5106	0.5179	0.4695	0.5694	0.4991	8.2384	**0.3686**	0.4345
AXP	0.2035	0.2807	0.2350	0.2013	0.1992	0.1616	0.3605	0.1884	16.9035	**0.1456**	0.1486
BA	31.7150	0.4817	0.3190	0.2305	0.2268	0.1941	0.4017	0.2226	15.5799	0.1695	**0.1651**
CAT	0.1398	0.1436	0.1792	0.1586	0.1626	0.1321	0.2276	0.1566	15.7173	**0.1210**	0.1268
CRM	0.2246	0.2228	0.2950	0.2551	0.2554	0.2227	0.2723	0.2501	15.0694	**0.1898**	0.2011
CSCO	0.9244	0.1683	0.2543	0.1986	0.1988	0.1596	0.2837	0.1922	20.0926	0.1490	**0.1477**
CVX	2.5162	0.2777	0.2162	0.1504	0.1523	0.1339	0.3232	0.1462	16.8956	**0.1246**	0.1253
DIS	8.5696	0.2826	0.2174	0.2182	0.2263	0.1704	0.3200	0.2161	16.3909	0.1657	**0.1616**
GE	0.1648	0.1627	0.2036	0.1908	0.1885	0.1653	0.2595	0.1862	12.3244	**0.1417**	0.1502
GOOGL	2.1229	0.2661	0.3177	0.3076	0.3215	0.2700	0.3457	0.3073	17.5032	**0.2292**	0.2476
GS	0.1475	0.1570	0.2097	0.1515	0.1544	0.1238	0.2465	0.1485	18.5384	0.1237	**0.1190**
HD	0.1990	0.2091	0.2376	0.1875	0.1869	0.1533	0.3327	0.1758	18.9064	**0.1384**	0.1399
HON	0.2820	0.3500	0.2460	0.2014	0.1968	0.1730	0.3479	0.1894	19.6790	0.1523	**0.1479**
IBM	1.6441	0.2038	0.2341	0.2044	0.2085	0.1654	0.2969	0.1952	18.1382	0.1535	**0.1471**
JNJ	0.2565	0.2141	0.2343	0.1922	0.1909	0.1935	0.2918	0.1856	20.3931	**0.1506**	0.1517
JPM	0.1760	0.1972	0.2489	0.1943	0.1959	0.1488	0.3191	0.1853	19.5637	**0.1426**	0.1434
KO	0.1661	0.1815	0.2126	0.1666	0.1670	0.2116	0.3091	0.1523	18.3551	0.1235	**0.1201**
MCD	5.7174	0.2857	0.2302	0.2107	0.2152	0.1992	0.3788	0.2024	20.2296	**0.1584**	0.1607
META	3.0672	0.1969	0.2529	0.1829	0.1842	0.1618	0.2561	0.1789	14.3516	**0.1399**	0.1423
MMM	0.6220	0.1950	0.2194	0.2026	0.2040	0.2049	0.2632	0.2021	18.9966	**0.1513**	0.1587
MRK	0.1776	0.1687	0.1848	0.1917	0.1898	0.1464	0.2338	0.1809	19.0289	**0.1372**	0.1417
MSFT	0.2953	0.2437	0.3282	0.2984	0.3080	0.2600	0.3749	0.2926	17.6576	**0.2221**	0.2397
NFLX	14.9187	0.3343	0.3298	0.3395	0.3433	0.3230	0.3636	0.3272	5.3863	**0.2632**	0.3027
NKE	0.2098	0.2113	0.2042	0.1934	0.1956	0.1476	0.2729	0.1826	16.7470	0.1487	**0.1442**
NVDA	0.1693	0.1642	0.2522	0.1836	0.1878	0.1700	0.2513	0.1880	3.8539	**0.1421**	0.1614
ORCL	4.6326	0.2788	0.3143	0.2984	0.3103	0.2443	0.3914	0.2904	15.6476	**0.2224**	0.2252
PG	0.2826	0.2035	0.1889	0.1569	0.1606	0.1858	0.2956	0.1510	19.5454	0.1227	**0.1189**
PM	0.2212	0.1903	0.2151	0.2176	0.2217	0.1803	0.2951	0.2186	21.1392	0.1739	**0.1715**
SHW	1.6041	0.2297	0.1700	0.1677	0.1711	0.1413	0.2595	0.1620	19.3818	0.1293	**0.1280**
TRV	0.1615	0.1726	0.1762	0.1513	0.1495	0.1431	0.2865	0.1452	17.7860	0.1184	**0.1171**
TSLA	0.1585	0.1437	0.2219	0.1931	0.1950	0.1746	0.2378	0.1907	1.6467	**0.1398**	0.1720
UNH	0.1972	0.1961	0.2286	0.2349	0.2334	0.1785	0.2883	0.2203	14.9049	**0.1640**	0.1676
V	1.1644	0.2749	0.2398	0.2532	0.2574	0.2197	0.3390	0.2466	18.2304	0.1970	**0.1936**
VZ	1.5620	0.1697	0.1933	0.1731	0.1814	0.1525	0.2606	0.1705	21.6736	**0.1366**	0.1400
WMT	0.9095	0.2674	0.2456	0.2355	0.2353	0.2188	0.2985	0.2345	21.7110	**0.1879**	0.1905
XOM	2.4919	0.3536	0.2074	0.1605	0.1594	0.1612	0.3179	0.1481	16.6871	0.1254	**0.1223**
Average	2.6979	0.2287	0.2513	0.2134	0.2160	0.1883	0.3048	0.2071	16.1397	**0.1613**	0.1673
FX											
AUDUSD	0.1179	0.1173	0.1485	0.1012	0.1039	0.2264	0.1806	0.1007	13.8959	**0.0820**	0.0934
EURUSD	0.1039	0.1003	0.1416	0.1005	0.1024	0.2346	0.1678	0.0999	10.0304	**0.0756**	0.1333
GBPUSD	0.1762	0.0998	0.1655	0.1167	0.1176	0.2573	0.1877	0.1155	10.8819	**0.0953**	0.1295
USDCAD	0.0914	0.0879	0.1142	0.0828	0.0844	0.1897	0.1487	0.0822	8.4382	**0.0669**	0.1188
USDJPY	0.2233	0.1953	0.3177	0.2375	0.2388	0.3421	0.2867	0.2319	11.5840	**0.1714**	0.2200
Average	0.1425	0.1201	0.1775	0.1277	0.1294	0.2500	0.1943	0.1260	10.9661	**0.0982**	0.1390
Futures											
C	0.1985	0.1772	0.2020	0.2159	0.2212	0.1851	0.2632	0.2100	17.2483	**0.1573**	0.1741
CL	0.2014	0.1645	0.2832	0.1672	0.1746	0.1772	0.4519	0.1430	7.5899	**0.1158**	0.1214
ES	1.4887	0.2531	0.4722	0.2805	0.2751	0.3454	0.5195	0.2618	19.5806	**0.2077**	0.2146
GC	0.1216	0.1057	0.2129	0.1297	0.1727	0.1952	0.1886	0.1279	18.8359	**0.1043**	0.1392
NG	0.1599	0.1231	0.1729	0.1322	0.1354	0.1301	0.2409	0.1332	1.3642	**0.1112**	0.1270
Average	0.4340	0.1647	0.2687	0.1851	0.1958	0.2066	0.3328	0.1752	12.9238	**0.1393**	0.1553
Overall Average	2.2160	0.2114	0.2457	0.2020	0.2053	0.1963	0.2965	0.1958	15.3007	**0.1528**	0.1633

Per-asset QLIKE loss at $h = 5$. QLIKE is computed as $\text{mean}(RV_t/\hat{RV}_t - \ln(RV_t/\hat{RV}_t) - 1)$. Bold indicates the lowest (best) QLIKE in each row. Chr-Bolt-S = Chronos-Bolt-Small, Chr-Bolt-B = Chronos-Bolt-Base, Moirai-S = Moirai-2.0-Small. HAR-J, HAR-RS, and HARQ are omitted as their QLIKE values are dominated by near-zero forecast artifacts (see Section 5.1).

Asset	HAR	Log-HAR	ARFIMA	Chr-Bolt-S	Chr-Bolt-B	Moirai-S	Lag-Llama	TFM-2.5	Toto	Sundial	MoE-S
Stocks											
AAPL	0.1969	**0.1808**	0.3361	0.3136	0.3193	0.2447	0.2470	0.3189	9.5453	0.1995	0.2180
ADBE	0.1398	**0.1348**	0.3791	0.2408	0.2470	0.2039	0.1963	0.2474	7.3768	0.1601	0.1721
AMD	0.0897	0.0884	0.1856	0.1324	0.1356	0.1319	0.1190	0.1314	0.1285	**0.0808**	0.1203
AMGN	3.2254	0.2634	**0.2177**	0.3563	0.3541	0.2988	0.2610	0.3605	3.4688	0.2416	0.2536
AMZN	23.0166	0.4877	0.4683	0.5492	0.5540	0.4759	0.4003	0.5499	2.7585	**0.2977**	0.4012
AXP	4.1452	0.3781	0.3191	0.3952	0.3945	0.2702	0.3868	0.3731	7.7764	**0.2497**	0.2561
BA	48.4755	1.1024	0.4859	0.4640	0.4634	0.4147	0.4528	0.4860	5.3179	**0.3265**	0.3316
CAT	4.6059	0.2131	0.1892	0.2444	0.2438	0.1883	0.1931	0.2436	7.7874	**0.1588**	0.1698
CRM	11.4276	0.3410	0.2683	0.2891	0.2881	0.2422	0.2096	0.2921	6.9779	**0.1761**	0.2044
CSCO	2.6118	0.2628	0.3505	0.2953	0.2982	0.2257	0.2550	0.3016	6.5477	**0.2048**	0.2149
CVX	16.7375	0.7459	0.3206	0.3418	0.3423	0.2794	0.3964	0.3527	11.6332	**0.2629**	0.2661
DIS	16.3347	0.5921	**0.1851**	0.3167	0.3324	0.2301	0.3033	0.3306	7.4460	0.2175	0.2111
GE	0.1934	0.2004	0.2111	0.2373	0.2380	0.1991	0.2203	0.2434	4.5332	**0.1542**	0.1665
GOOGL	6.7952	0.2986	0.2545	0.3646	0.3818	0.3120	0.2549	0.3959	4.3320	**0.2299**	0.2677
GS	0.2048	0.2145	0.2868	0.2503	0.2558	0.2007	0.2303	0.2576	6.0486	0.1974	**0.1890**
HD	1.4765	0.3321	0.3579	0.3579	0.3567	0.2690	0.3500	0.3536	8.1227	**0.2459**	0.2614
HON	6.7514	0.5217	0.3450	0.3785	0.3761	0.2563	0.3582	0.3561	9.7074	0.2575	**0.2470**
IBM	2.1795	0.2645	0.2330	0.3178	0.3269	0.2153	0.2770	0.3070	6.6203	0.2144	**0.1963**
JNJ	0.3284	0.2763	0.2872	0.3486	0.3507	0.2890	0.2817	0.3434	8.2663	**0.2423**	0.2620
JPM	6.3848	0.3178	0.3438	0.3719	0.3691	0.2749	0.3335	0.3726	7.6711	**0.2447**	0.2633
KO	3.2457	0.3462	0.3343	0.3918	0.3967	0.2948	0.3511	0.3699	9.4222	0.2776	**0.2696**
MCD	4.4248	0.4672	0.3247	0.4223	0.4230	0.3013	0.4126	0.4320	9.6505	**0.2971**	0.3124
META	4.1675	0.2438	0.2527	0.2389	0.2411	0.2123	0.2056	0.2614	5.3226	**0.1611**	0.1746
MMM	1.9908	0.2210	0.2090	0.2714	0.2690	0.2264	0.2122	0.2807	6.7874	**0.1814**	0.1825
MRK	0.2130	0.1894	0.1718	0.2568	0.2582	0.1788	0.2079	0.2568	5.9427	**0.1692**	0.1750
MSFT	4.6384	0.2857	0.3099	0.3927	0.4017	0.3111	0.3071	0.4059	7.1386	**0.2430**	0.2696
NFLX	9.5100	0.3088	0.2297	0.3269	0.3278	0.3115	0.2546	0.3337	0.8458	**0.2161**	0.2783
NKE	2.9300	0.3044	0.2201	0.2841	0.2867	0.2167	0.2653	0.2804	7.7602	**0.2044**	0.2054
NVDA	0.2246	0.2052	0.2805	0.2322	0.2359	0.2086	0.2019	0.2497	0.6064	**0.1476**	0.1842
ORCL	7.1831	0.3691	0.3904	0.4655	0.4867	0.3851	0.3491	0.4817	7.3199	**0.2941**	0.3491
PG	3.1120	0.3395	0.3016	0.3307	0.3344	0.2721	0.3286	0.3210	9.5718	0.2466	**0.2400**
PM	0.2639	**0.2057**	0.2625	0.2900	0.2941	0.2525	0.2789	0.3214	5.5608	0.2172	0.2346
SHW	5.1294	0.2928	**0.1803**	0.2545	0.2588	0.2181	0.2464	0.2674	7.5363	0.1836	0.1946
TRV	0.2825	0.3321	0.2575	0.2930	0.2850	0.2293	0.2953	0.2972	5.0933	**0.2172**	0.2295
TSLA	0.1561	0.1660	0.1848	0.2469	0.2520	0.2213	0.1921	0.2502	0.1654	**0.1536**	0.2004
UNH	3.1724	0.2263	0.2006	0.3292	0.3209	0.2221	0.2567	0.3117	5.3770	0.2072	**0.1960**
V	3.8648	0.4143	0.2704	0.3721	0.3807	0.2710	0.3090	0.3743	8.5910	0.2505	**0.2494**
VZ	1.9827	0.2200	0.2335	0.2702	0.2756	0.2068	0.2390	0.2798	6.3886	**0.1956**	0.2097
WMT	6.0957	0.4762	0.2769	0.3359	0.3330	0.2550	0.2697	0.3309	9.1863	**0.2321**	0.2451
XOM	19.0392	0.9555	0.3003	0.3194	0.3247	0.2565	0.3941	0.3190	7.2077	0.2479	**0.2343**
Average	5.9237	0.3446	0.2804	0.3223	0.3253	0.2568	0.2826	0.3261	6.4385	**0.2176**	0.2327
FX											
AUDUSD	0.1665	0.1624	0.1645	0.1441	0.1478	0.2003	0.1739	0.1467	10.7784	**0.1081**	0.1151
EURUSD	0.1267	0.1306	0.1469	0.1232	0.1284	0.2007	0.1409	0.1232	8.4332	**0.0850**	0.1284
GBPUSD	0.2949	0.1254	0.1719	0.1479	0.1485	0.2240	0.1533	0.1514	9.7204	**0.1133**	0.1515
USDCAD	0.1115	0.1092	0.1240	0.1158	0.1196	0.1637	0.1329	0.1172	7.3388	**0.0847**	0.1123
USDJPY	0.2295	0.2081	0.3196	0.2741	0.2714	0.2701	0.2166	0.2852	9.8760	**0.1690**	0.2157
Average	0.1858	0.1471	0.1854	0.1610	0.1631	0.2117	0.1635	0.1647	9.2293	**0.1120**	0.1446
Futures											
C	0.1695	0.1435	0.1502	0.2023	0.2040	0.1563	0.1853	0.1933	7.1465	**0.1176**	0.1342
CL	0.3207	0.4053	0.4021	0.2964	0.2970	0.2697	0.5906	0.2564	2.6843	**0.1854**	0.1854
ES	4.5906	0.6009	0.5021	0.5946	0.6009	0.4383	0.5478	0.5986	12.4722	**0.3753**	0.4187
GC	0.1340	**0.1209**	0.2085	0.1683	0.1770	0.1760	0.1553	0.1691	9.3199	0.1262	0.1516
NG	0.1661	0.1317	0.1653	0.1477	0.1520	0.1478	0.2034	0.1584	0.2368	**0.1152**	0.1377
Average	1.0762	0.2805	0.2857	0.2819	0.2862	0.2376	0.3365	0.2752	6.3719	**0.1839**	0.2055
Overall Average	4.8651	0.3185	0.2714	0.3021	0.3052	0.2504	0.2761	0.3048	6.7109	**0.2037**	0.2212

Per-asset QLIKE loss at $h = 22$. QLIKE is computed as $\text{mean}(RV_t/\hat{RV}_t - \ln(RV_t/\hat{RV}_t) - 1)$. Bold indicates the lowest (best) QLIKE in each row. Chr-Bolt-S = Chronos-Bolt-Small, Chr-Bolt-B = Chronos-Bolt-Base, Moirai-S = Moirai-2.0-Small. HAR-J, HAR-RS, and HARQ are omitted as their QLIKE values are dominated by near-zero forecast artifacts (see Section 5.1).

Table 17: Mincer–Zarnowitz forecast efficiency regression ($h = 5$). See Table 11 for variable definitions.

Model	$\hat{\alpha}$	$\hat{\beta}$	R^2	F-stat	p(joint)	% Reject
Econometric benchmarks						
HAR	0.0000	1.155	0.276	7.09	0.261	28%
HAR-J	0.0000	1.154	0.293	5.99	0.268	28%
HAR-RS	0.0001	0.910	0.210	15.64	0.143	52%
HARQ	0.0001	0.681	0.119	13.01	0.063	70%
Log-HAR	−0.0000	1.438	0.198	5.42	0.050	76%
ARFIMA	0.0001	0.977	0.221	15.90	0.099	72%
Foundation models						
Chronos-Bolt-S	0.0000	1.282	0.400	26.29	0.002	98%
Chronos-Bolt-B	0.0000	1.438	0.417	31.24	0.001	98%
Moirai-2.0-S	0.0000	1.094	0.367	18.12	0.007	94%
Lag-Llama	0.0000	1.183	0.069	5.01	0.055	60%
TimesFM-2.5	0.0000	1.324	0.437	27.42	0.001	98%
Toto	0.0001	0.483	0.337	23.14	0.006	98%
Sundial	0.0000	1.136	0.405	11.80	0.035	82%
Moirai-MoE-S	0.0001	0.751	0.444	22.10	0.001	98%

Table 18: Mincer–Zarnowitz forecast efficiency regression ($h = 22$). See Table 11 for variable definitions.

Model	$\hat{\alpha}$	$\hat{\beta}$	R^2	F-stat	p(joint)	% Reject
Econometric benchmarks						
HAR	0.0001	0.490	0.096	18.82	0.044	70%
HAR-J	0.0001	0.510	0.102	16.83	0.044	70%
HAR-RS	0.0002	0.366	0.083	32.16	0.016	92%
HARQ	0.0002	0.299	0.074	54.19	0.003	100%
Log-HAR	0.0001	0.474	0.072	16.70	0.018	92%
ARFIMA	0.0001	0.578	0.143	16.87	0.040	80%
Foundation models						
Chronos-Bolt-S	0.0001	1.121	0.202	10.98	0.008	96%
Chronos-Bolt-B	0.0000	1.453	0.221	12.15	0.005	98%
Moirai-2.0-S	0.0000	0.985	0.187	12.55	0.006	98%
Lag-Llama	0.0001	0.705	0.057	3.43	0.096	50%
TimesFM-2.5	0.0000	1.301	0.235	12.74	0.005	98%
Toto	0.0002	0.255	0.172	28.87	0.009	98%
Sundial	0.0001	0.892	0.211	5.88	0.043	80%
Moirai-MoE-S	0.0001	0.483	0.203	16.79	0.004	98%

Table 19: Estimation window sensitivity ($h = 5$). Estimation window sensitivity: 252-day vs. 512-day rolling window ($h = 5$, 40 equities). Cross-asset mean metrics. Bold indicates the better window for each model.

	MSE ($\times 10^8$)		QLIKE		R^2_{OOS}	
Model	252	512	252	512	252	512
HAR	14.95	**12.32**	2.4626	**0.1794**	0.221	**0.389**
HAR-J	14.65	**11.90**	2.2567	**0.2355**	0.242	**0.407**
HAR-RS	17.09	**12.84**	6.3478	**0.4547**	0.108	**0.360**
HARQ	17.54	**14.45**	14.4245	**4.7712**	0.040	**0.240**
Log-HAR	14.94	**13.38**	0.2211	**0.1628**	0.156	**0.316**
ARFIMA	**16.37**	29.68	**0.2551**	0.4191	**0.144**	-0.659

Table 20: Estimation window sensitivity ($h = 22$). Estimation window sensitivity: 252-day vs. 512-day rolling window ($h = 22$, 40 equities). Cross-asset mean metrics. Bold indicates the better window for each model.

	MSE ($\times 10^8$)		QLIKE		R^2_{OOS}	
Model	252	512	252	512	252	512
HAR	10.71	**9.33**	5.3791	**0.2744**	-0.011	**0.145**
HAR-J	10.66	**9.25**	5.2619	**0.3315**	0.004	**0.154**
HAR-RS	11.88	**9.98**	8.4756	**0.7274**	-0.126	**0.089**
HARQ	11.95	**10.05**	10.9013	**3.3530**	-0.158	**0.066**
Log-HAR	10.42	**9.29**	0.3247	**0.1966**	-0.017	**0.149**
ARFIMA	**11.47**	26.51	**0.2757**	0.4123	**-0.058**	-1.564